

Distribution of Prime Numbers on the Radial Layer Spiral (RLS)

Ryszard Waluś

`richi.wales@gmail.com`

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Abstract

The paper presents a geometric representation of natural numbers based on the Radial Layer Spiral (RLS), in which each natural number is uniquely assigned to a cell of a square grid determined algorithmically.

In this representation, the distribution of prime numbers reveals a visually ordered and symmetric structure over a wide range of values.

In contrast to classical asymptotic results, such as the Prime Number Theorem, which describe only the average density, the RLS framework reveals the spatial organization of the distribution of prime numbers.

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1 Introduction

We consider an infinite square grid of cells. A distinguished central cell is selected, whose center is treated as the starting point.

Assumptions

1. We define a path line as a curve that passes exactly once through the centers of the grid cells, thereby determining their ordering.
2. The path line uniquely assigns consecutive natural numbers to the grid cells through whose centers it passes.
3. The set of cells whose centers are traversed by the path line up to a given step forms a connected region.
4. This region expands monotonically and tends to cover the entire grid.

Let A_n denote the set of cells whose centers are traversed by the path line after n steps. Then

$$A_1 \subset A_2 \subset \dots, \quad \bigcup_{n=1}^{\infty} A_n = \mathbb{Z}^2,$$

and each set A_n is connected.

This construction defines a mapping

$$f: \mathbb{N} \rightarrow \mathbb{Z}^2,$$

which assigns to each natural number n exactly one cell of the grid, identified by the index pair (i, j) .

This mapping is unique, that is,

$$\forall n \in \mathbb{N} \quad \exists! (i, j) \in \mathbb{Z}^2 \quad \text{such that} \quad f(n) = (i, j).$$

In the following sections, two example paths on the square grid and the corresponding distributions of prime numbers will be presented. Subsequently, the Radial Layer Spiral (RLS) will be introduced, which incorporates both the layered and radial structure with respect to the central point.

Example Paths on the Square Grid

Figure 1 shows the scheme of the Ulam spiral [2], and Figure 2 presents the corresponding distribution of prime numbers.

Figure 3 shows the scheme of the reverse spiral, and Figure 4 presents the corresponding distribution of prime numbers.

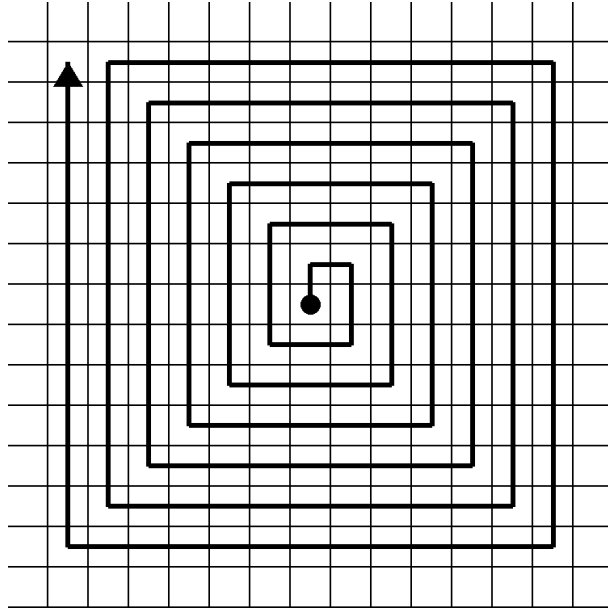


Figure 1: Scheme of the spiral path corresponding to the Ulam spiral.

842	843	844	845	846	847	848	849	850	851	852	853	854	855	856	857	858	859	860	861	862	863	864	865	866	867	868	869	870	871
841	730	731	732	733	734	735	736	737	738	739	740	741	742	743	744	745	746	747	748	749	750	751	752	753	754	755	756	757	872
840	729	626	627	628	629	630	631	632	633	634	635	636	637	638	639	640	641	642	643	644	645	646	647	648	649	650	651	758	873
839	728	625	530	531	532	533	534	535	536	537	538	539	540	541	542	543	544	545	546	547	548	549	550	551	552	553	652	759	874
838	727	624	529	442	443	444	445	446	447	448	449	450	451	452	453	454	455	456	457	458	459	460	461	462	463	554	653	760	875
837	726	623	528	441	362	363	364	365	366	367	368	369	370	371	372	373	374	375	376	377	378	379	380	381	464	555	654	761	876
836	725	622	527	440	361	290	291	292	293	294	295	296	297	298	299	300	301	302	303	304	305	306	307	382	465	556	655	762	877
835	724	621	526	439	360	289	226	227	228	229	230	231	232	233	234	235	236	237	238	239	240	241	308	383	466	557	656	763	878
834	723	620	525	438	359	288	225	170	171	172	173	174	175	176	177	178	179	180	181	182	183	242	309	384	467	558	657	764	879
833	722	619	524	437	358	287	224	169	122	123	124	125	126	127	128	129	130	131	132	133	184	243	310	385	468	559	658	765	880
832	721	618	523	436	357	286	223	168	121	82	83	84	85	86	87	88	89	90	91	134	185	244	311	386	469	560	659	766	881
831	720	617	522	435	356	285	222	167	120	81	50	51	52	53	54	55	56	57	92	135	186	245	312	387	470	561	660	767	882
830	719	616	521	434	355	284	221	166	119	80	49	26	27	28	29	30	31	58	93	136	187	246	313	388	471	562	661	768	883
829	718	615	520	433	354	283	220	165	118	79	48	25	10	11	12	13	32	59	94	137	188	247	314	389	472	563	662	769	884
828	717	614	519	432	353	282	219	164	117	78	47	24	9	2	3	14	33	60	95	138	189	248	315	390	473	564	663	770	885
827	716	613	518	431	352	281	218	163	116	77	46	23	8	1	4	15	34	61	96	139	190	249	316	391	474	565	664	771	886
826	715	612	517	430	351	280	217	162	115	76	45	22	7	6	5	16	35	62	97	140	191	250	317	392	475	566	665	772	887
825	714	611	516	429	350	279	216	161	114	75	44	21	20	19	18	17	36	63	98	141	192	251	318	393	476	567	666	773	888
824	713	610	515	428	349	278	215	160	113	74	43	42	41	40	39	38	37	64	99	142	193	252	319	394	477	568	667	774	889
823	712	609	514	427	348	277	214	159	112	73	72	71	70	69	68	67	66	65	100	143	194	253	320	395	478	569	668	775	890
822	711	608	513	426	347	276	213	158	111	110	109	108	107	106	105	104	103	102	101	144	195	254	321	396	479	570	669	776	891
821	710	607	512	425	346	275	212	157	156	155	154	153	152	151	150	149	148	147	146	145	196	255	322	397	480	571	670	777	892
820	709	606	511	424	345	274	211	210	209	208	207	206	205	204	203	202	201	200	199	198	197	256	323	398	481	572	671	778	893
819	708	605	510	423	344	273	272	271	270	269	268	267	266	265	264	263	262	261	260	259	258	257	324	399	482	573	672	779	894
818	707	604	509	422	343	342	341	340	339	338	337	336	335	334	333	332	331	330	329	328	327	326	325	400	483	574	673	780	895
817	706	603	508	421	420	419	418	417	416	415	414	413	412	411	410	409	408	407	406	405	404	403	402	401	484	575	674	781	896
816	705	602	507	506	505	504	503	502	501	500	499	498	497	496	495	494	493	492	491	490	489	488	487	486	485	576	675	782	897
815	704	601	600	599	598	597	596	595	594	593	592	591	590	589	588	587	586	585	584	583	582	581	580	579	578	577	676	783	898
814	703	702	701	700	699	698	697	696	695	694	693	692	691	690	689	688	687	686	685	684	683	682	681	680	679	678	677	784	899
813	812	811	810	809	808	807	806	805	804	803	802	801	800	799	798	797	796	795	794	793	792	791	790	789	788	787	786	785	900

Figure 2: Distribution of prime numbers for the Ulam spiral.

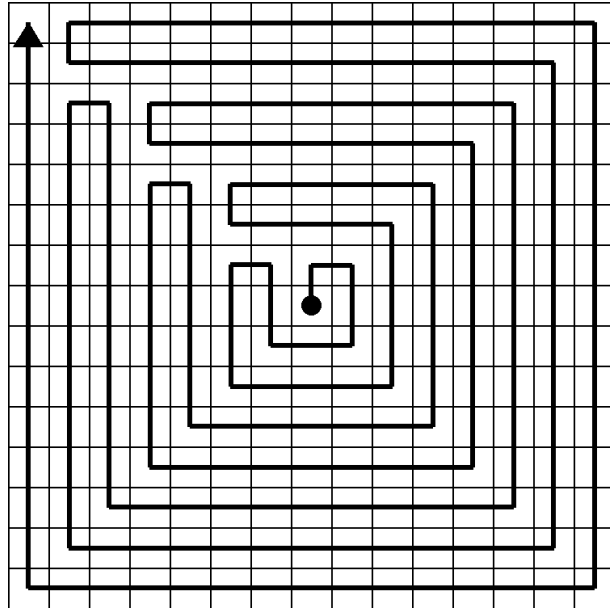


Figure 3: Scheme of the reverse spiral path.

730	729	626	627	628	629	630	631	632	633	634	635	636	637	638	639	640	641	642	643	644	645	646	647	648	649	650	651
731	728	625	624	623	622	621	620	619	618	617	616	615	614	613	612	611	610	609	608	607	606	605	604	603	602	601	652
732	727	530	529	442	443	444	445	446	447	448	449	450	451	452	453	454	455	456	457	458	459	460	461	462	463	600	653
733	726	531	528	441	440	439	438	437	436	435	434	433	432	431	430	429	428	427	426	425	424	423	422	421	464	599	654
734	725	532	527	362	361	290	291	292	293	294	295	296	297	298	299	300	301	302	303	304	305	306	307	420	465	598	655
735	724	533	526	363	360	289	288	287	286	285	284	283	282	281	280	279	278	277	276	275	274	273	308	419	466	597	656
736	723	534	525	364	359	226	225	170	171	172	173	174	175	176	177	178	179	180	181	182	183	272	309	418	467	596	657
737	722	535	524	365	358	227	224	169	168	167	166	165	164	163	162	161	160	159	158	157	184	271	310	417	468	595	658
738	721	536	523	366	357	228	223	122	121	82	83	84	85	86	87	88	89	90	91	156	185	270	311	416	469	594	659
739	720	537	522	367	356	229	222	123	120	81	80	79	78	77	76	75	74	73	92	155	186	269	312	415	470	593	660
740	719	538	521	368	355	230	221	124	119	50	49	26	27	28	29	30	31	72	93	154	187	268	313	414	471	592	661
741	718	539	520	369	354	231	220	125	118	51	48	25	24	23	22	21	32	71	94	153	188	267	314	413	472	591	662
742	717	540	519	370	353	232	219	126	117	52	47	10	9	2	3	20	33	70	95	152	189	266	315	412	473	590	663
743	716	541	518	371	352	233	218	127	116	53	46	11	8	1	4	19	34	69	96	151	190	265	316	411	474	589	664
744	715	542	517	372	351	234	217	128	115	54	45	12	7	6	5	18	35	68	97	150	191	264	317	410	475	588	665
745	714	543	516	373	350	235	216	129	114	55	44	13	14	15	16	17	36	67	98	149	192	263	318	409	476	587	666
746	713	544	515	374	349	236	215	130	113	56	43	42	41	40	39	38	37	66	99	148	193	262	319	408	477	586	667
747	712	545	514	375	348	237	214	131	112	57	58	59	60	61	62	63	64	65	100	147	194	261	320	407	478	585	668
748	711	546	513	376	347	238	213	132	111	110	109	108	107	106	105	104	103	102	101	146	195	260	321	406	479	584	669
749	710	547	512	377	346	239	212	133	134	135	136	137	138	139	140	141	142	143	144	145	196	259	322	405	480	583	670
750	709	548	511	378	345	240	211	210	209	208	207	206	205	204	203	202	201	200	199	198	197	258	323	404	481	582	671
751	708	549	510	379	344	241	242	243	244	245	246	247	248	249	250	251	252	253	254	255	256	257	324	403	482	581	672
752	707	550	509	380	343	342	341	340	339	338	337	336	335	334	333	332	331	330	329	328	327	326	325	402	483	580	673
753	706	551	508	381	382	383	384	385	386	387	388	389	390	391	392	393	394	395	396	397	398	399	400	401	484	579	674
754	705	552	507	506	505	504	503	502	501	500	499	498	497	496	495	494	493	492	491	490	489	488	487	486	485	578	675
755	704	553	554	555	556	557	558	559	560	561	562	563	564	565	566	567	568	569	570	571	572	573	574	575	576	577	676
756	703	702	701	700	699	698	697	696	695	694	693	692	691	690	689	688	687	686	685	684	683	682	681	680	679	678	677
757	758	759	760	761	762	763	764	765	766	767	768	769	770	771	772	773	774	775	776	777	778	779	780	781	782	783	784

Figure 4: Distribution of prime numbers for the reverse spiral.

2 Radial Layer Spiral (RLS): Construction

The starting point is the analysis of the structure of layers propagating radially from the central cell on the square grid.

This leads to the formulation of the concept of a path line, in which consecutive natural numbers are assigned to cells belonging to successive radial layers.

These layers define a natural partition of the grid into sets of cells with increasing distance from the central point.

For further description, we introduce the following notation for quantities and relations:

- $R = \frac{a}{2} = 1$ — radius of the circle,
- $a = 2R$ — side length of the square in which the circle is inscribed,
- b — side length of a small grid cell,
- $c = \frac{a}{b}$ — number of subdivisions of the grid along one side (grid density),
- N — number of cell centers lying inside the circle of radius R ,
- r_p — radius of a radial layer, i.e., the distance of the center of a cell belonging to a given layer from the origin.

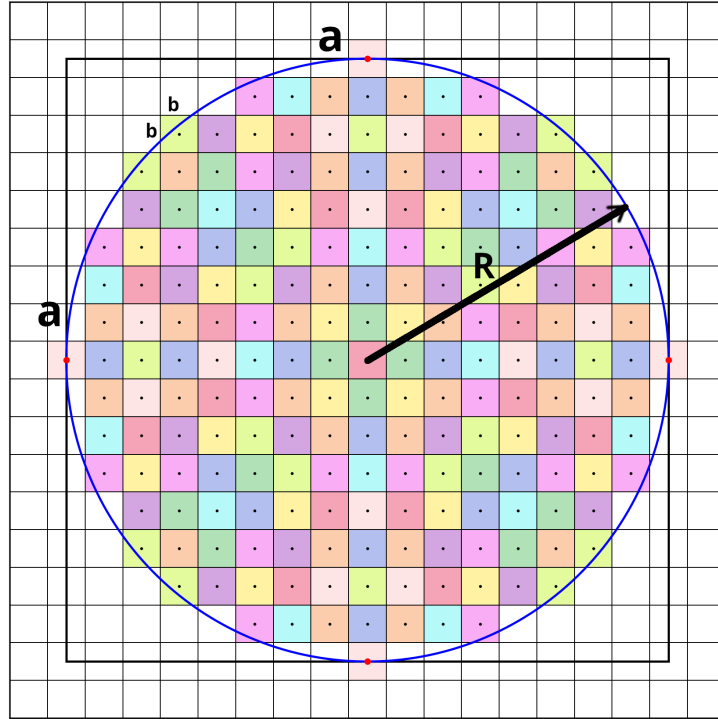


Figure 5: A square grid with step b overlaid on a circle of radius R inscribed in a square of side length $a = 2R$. The centers of the grid cells define a discrete set of points on the basis of which radial layers are defined. The number of subdivisions of the grid along one side is $c = \frac{a}{b}$. The set of cell centers lying inside the circle contains N points and forms a discrete approximation of the circular domain.

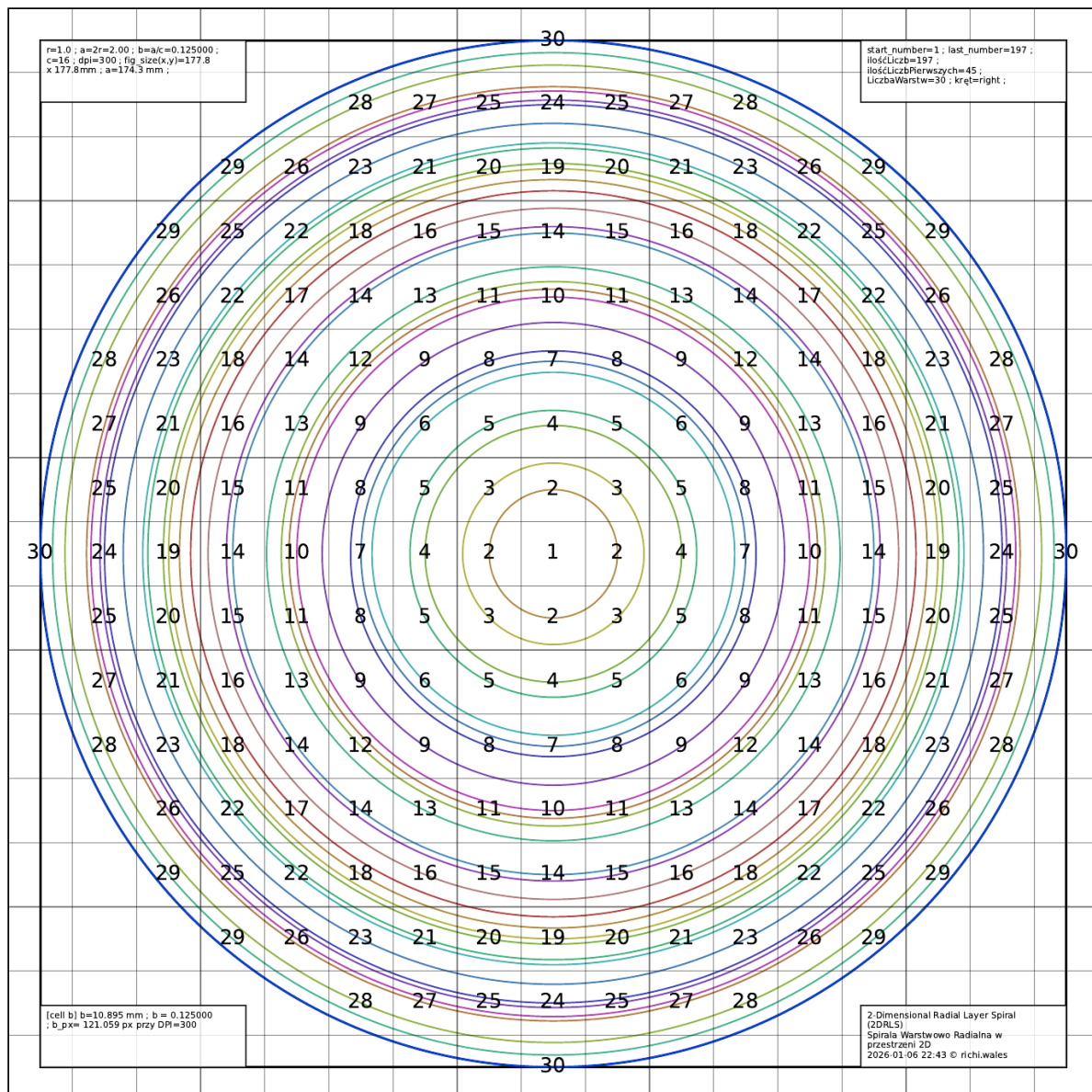


Figure 6: Circles with radii corresponding to successive layers pass through the centers of the cells belonging to a given layer. Cells of each layer are labeled by the layer index. Number of layers: 30.

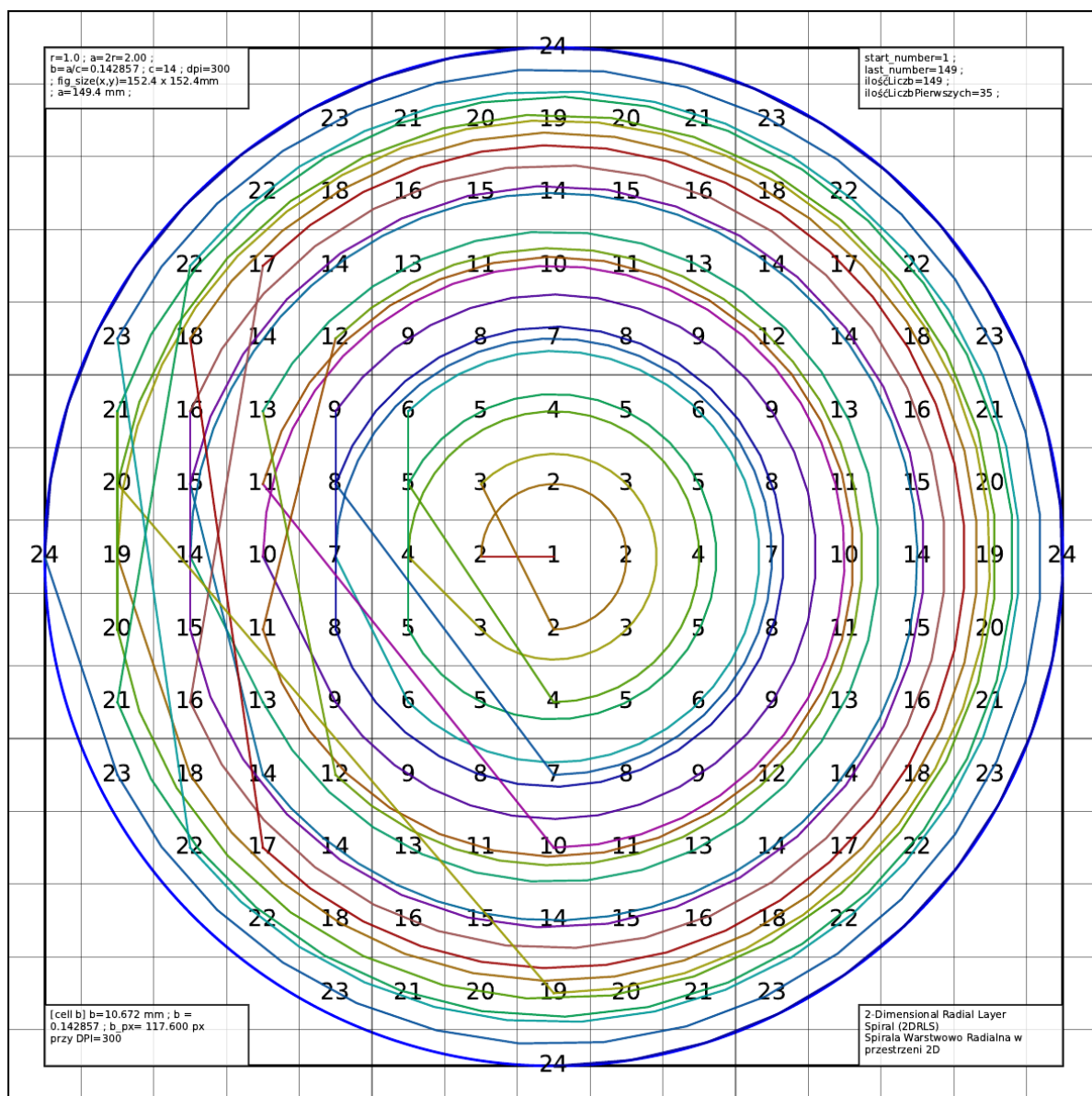


Figure 7: A spiral passing through successive layers. Cells are labeled by their corresponding layer index. Number of layers: 24.

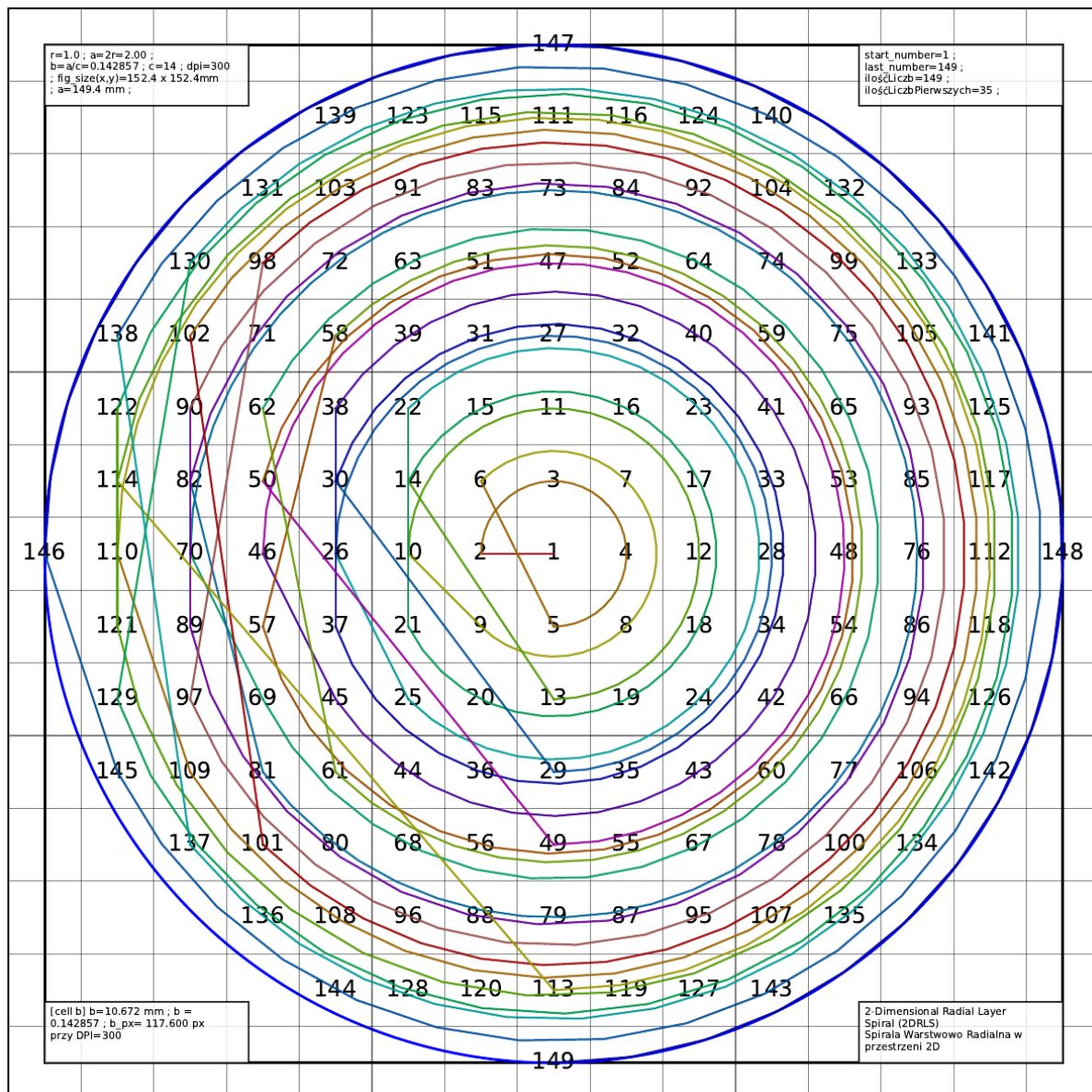


Figure 8: A schematic showing the path line of the Radial Layer Spiral (RLS) passing through successive cells of the layers, labeled by consecutive natural numbers from 1 to 149. Number of layers: 24. Number of cells: 149.

Figures 6 and 8 illustrate two key aspects of the RLS construction. The first shows the structure of radial layers determined by circles with increasing radii r_p , which pass through the centers of the grid cells.

The second figure presents the method of ordering the cells by means of the path line, which assigns consecutive natural numbers to cells belonging to successive radial layers.

In this way, we obtain an ordering of the grid cells that follows directly from the structure of radial layers. This construction is called the Radial Layer Spiral (RLS), introduced in [1].

Figure 9 shows the arrangement of natural numbers in the RLS construction. Cells corresponding to prime numbers are highlighted in black.

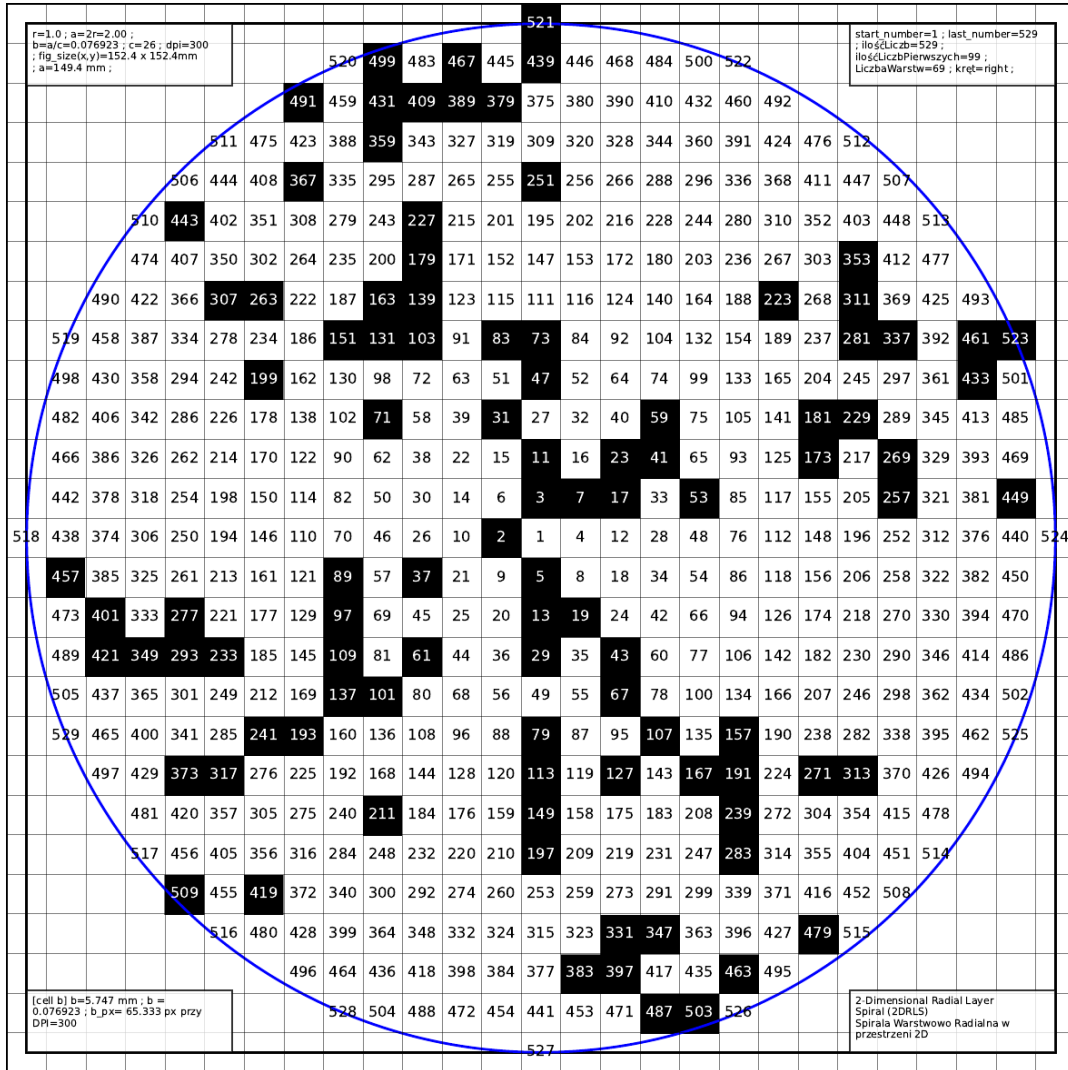


Figure 9: Consecutive natural numbers from 1 to 529 arranged according to the RLS spiral. Black cells indicate prime numbers. Figure parameters: number of cells = 529; number of layers = 69; number of grid subdivisions $c = 26$; number of prime numbers = 99.

In the subsequent figures, increasingly finer grids of cells are used, allowing visualization of progressively larger ranges of numbers. Figure 10 presents the range 1–70 681, in which the number of prime numbers is 7 002. Figure 11 shows the range 1–785 349, containing 62 869 prime numbers. In turn, Figure 12 presents the range 1–7 068 569, for which the number of prime

numbers is 481 004.

For smaller ranges, a clear and symmetric structure in the distribution of prime numbers is observed, whose structure resembles a four-bladed propeller (see Figure 10). As the range of analyzed numbers increases, a pattern emerges with a structure similar to an eight-bladed propeller (see Figures 11 and 12).

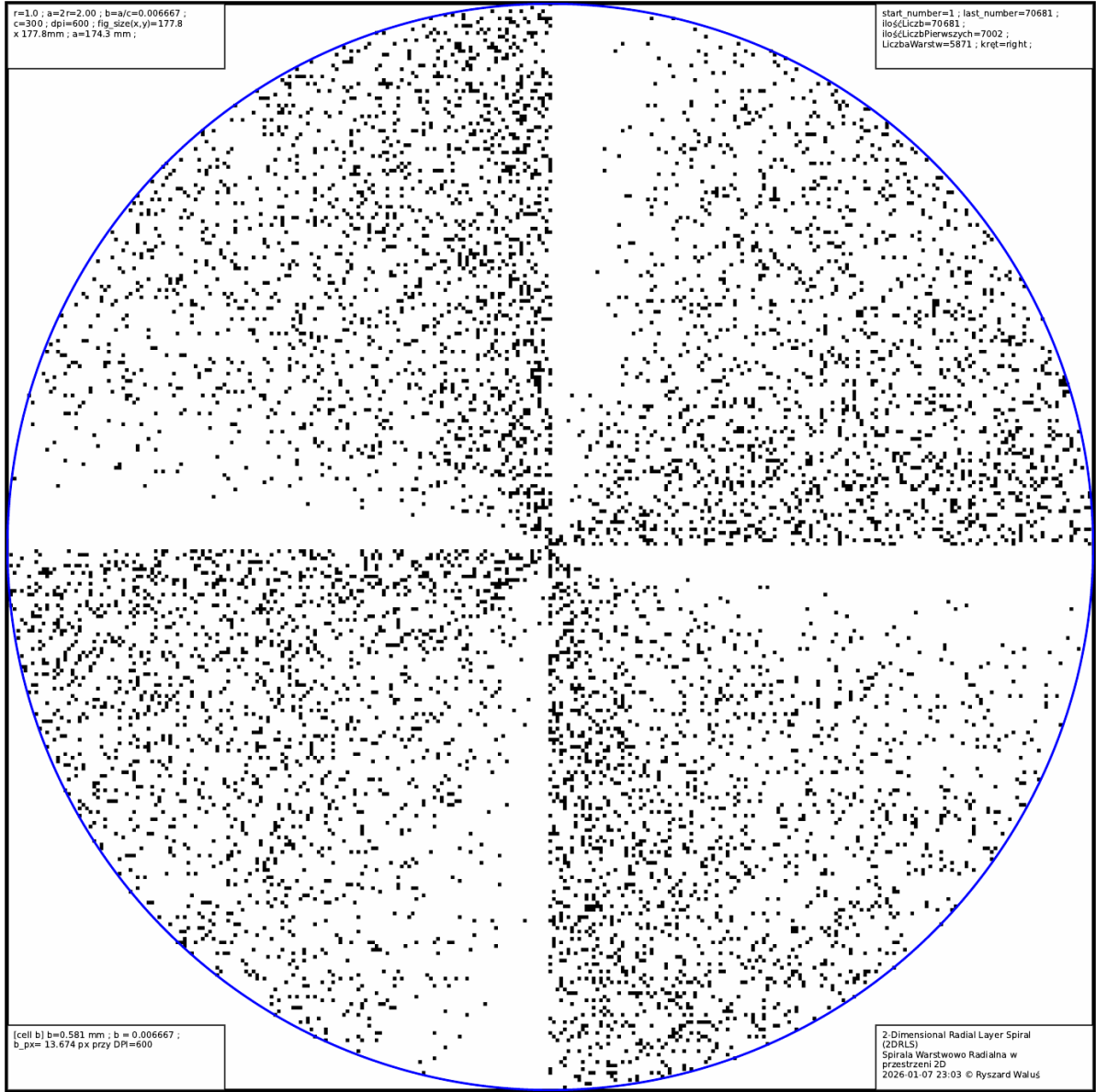


Figure 10: Distribution of prime numbers on the RLS spiral. Figure parameters: number of grid subdivisions $c = 300$; range of numbers 1–70 681; number of cells = 70 681; number of layers = 5 871; number of prime numbers = 7 002.

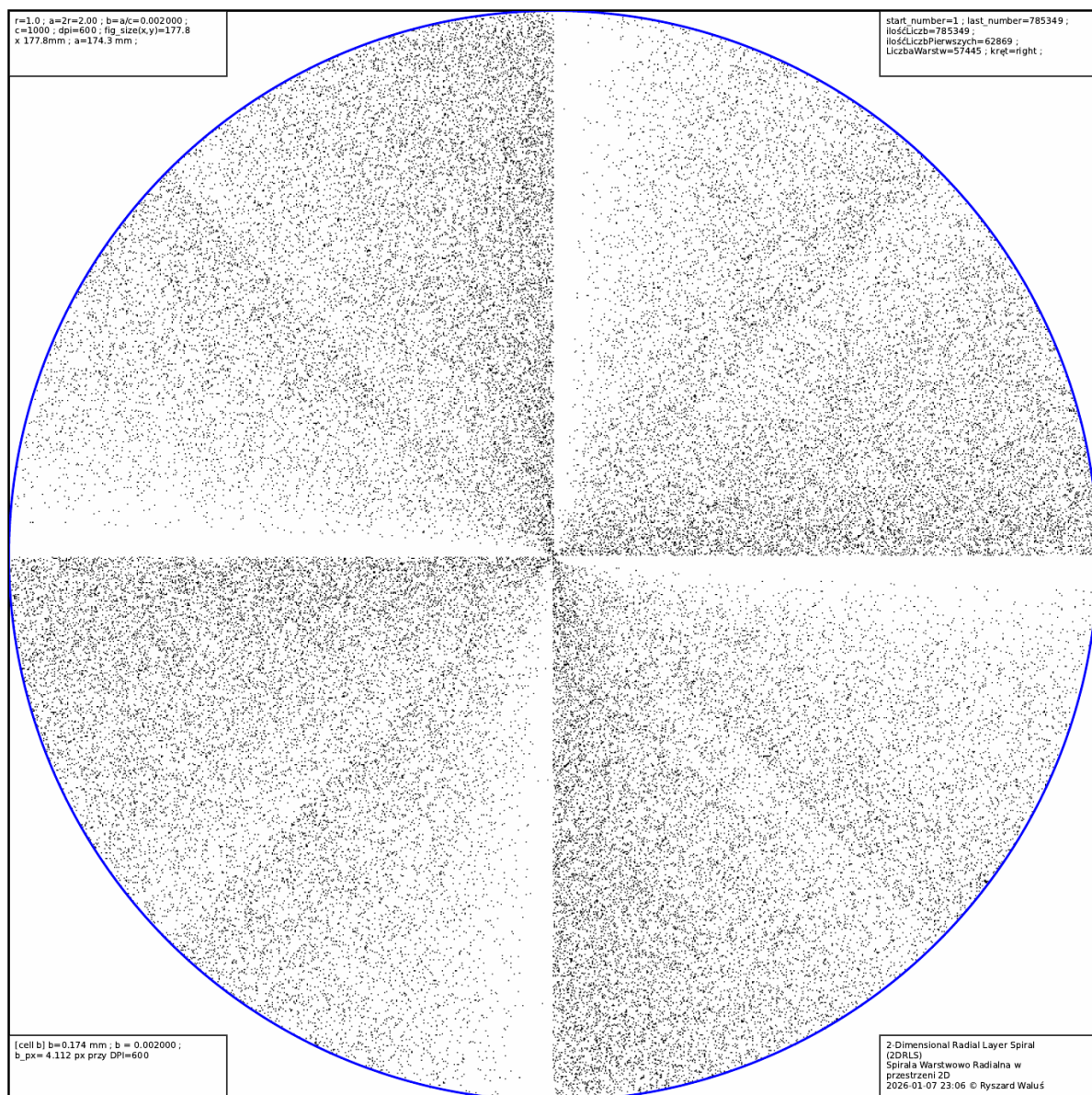


Figure 11: Distribution of prime numbers on the RLS spiral. Figure parameters: number of grid subdivisions $c = 1000$; range of numbers 1–785 349; number of cells = 785 349; number of layers = 57 445; number of prime numbers = 62 869.

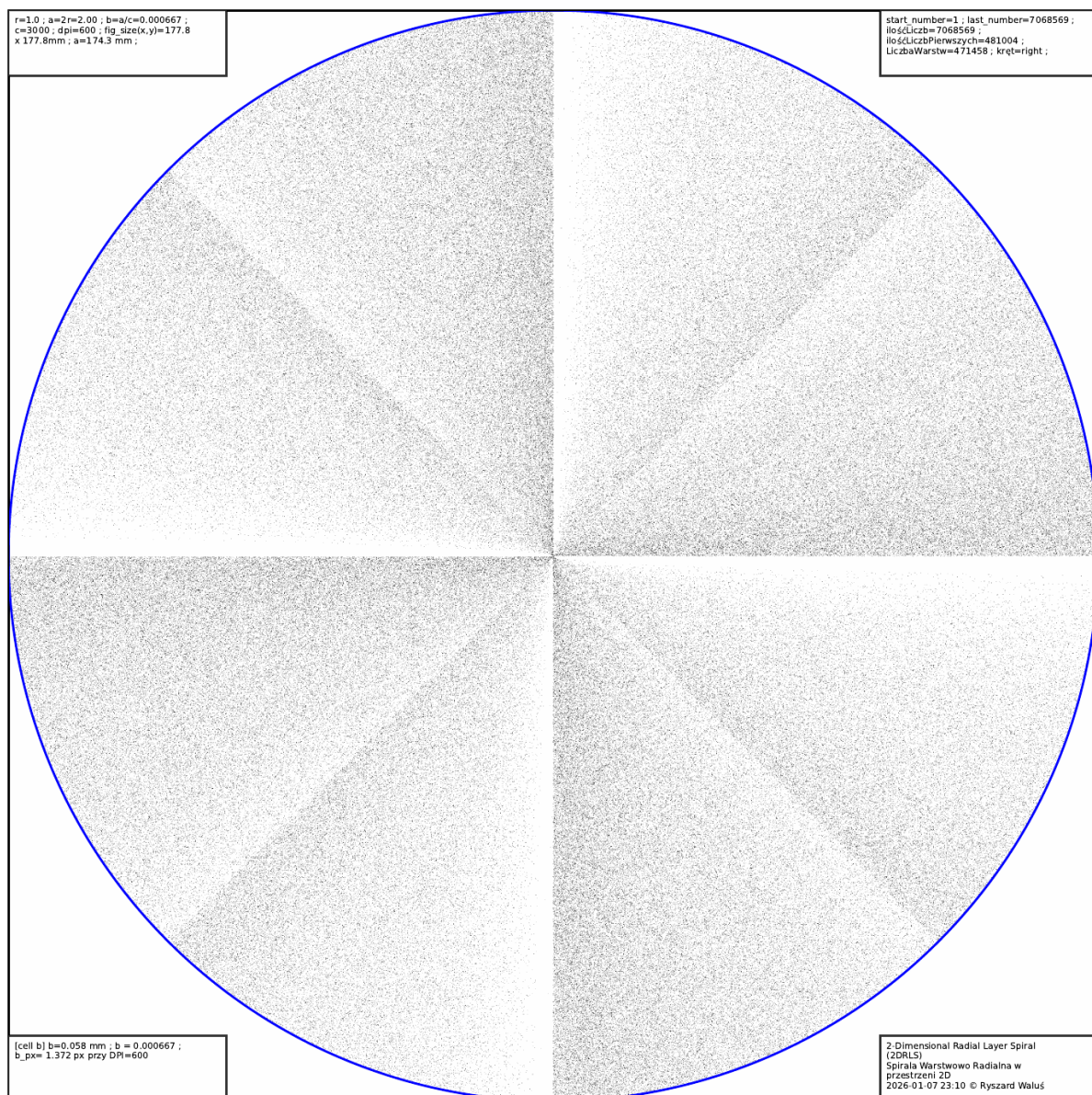


Figure 12: Distribution of prime numbers on the RLS spiral. Figure parameters: number of grid subdivisions $c = 3000$; range of numbers 1–7 068 569; number of cells = 7 068 569; number of layers = 471 458; number of prime numbers = 481 004.

3 Analysis of the Distribution of Prime Numbers on the Radial Layer Spiral (RLS)

Partition of the Circle into Angular Sectors

In order to determine the density distribution of prime numbers, the circle is divided into angular sectors. Figure 13 illustrates, for reference, a circle divided into 36 equal sectors, each representing a circular sector with an angle of 10° .

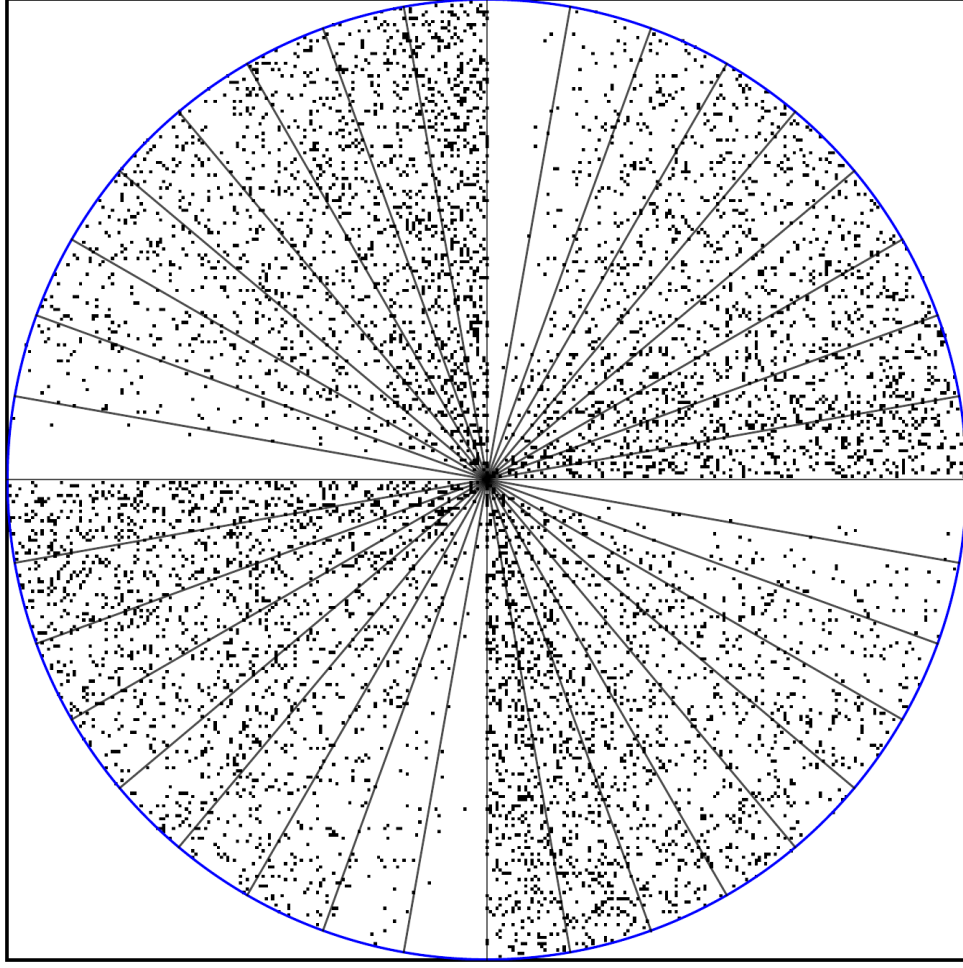


Figure 13: Number of grid subdivisions $c = 300$; number of cells: 70 681; number of layers: 5 871; number of prime numbers: 7 002; number of angular sectors: 36. Each sector represents a circular sector with an angle of 10° .

The basic set of data assigned to each cell includes:

- the cell coordinates in the square grid (i, j) ,
- the coordinates of the cell center in the Cartesian coordinate system (x, y) , scaled to a circle of radius $R = 1$,
- the radial distance of the cell center r_p (denoted in the database as `r_p`),
- the natural number n assigned to the given cell,
- the layer index (`layer`) to which the cell belongs,
- the flag `is_prime`, indicating whether the number n is a prime number.

The collected data enable the determination of a range of derived quantities, in particular the polar coordinates (r, θ) , where r denotes the distance of the point from the origin, while θ represents the angle between the positive direction of the x -axis and the vector connecting the origin with the considered point, expressed in radians or degrees.

The cell data were computed based on a square grid with parameter $c = 7500$, yielding a total of 44 178 653 cells contained within the circle. In the analyzed range, 2 671 482 prime numbers were identified, and the number of spiral layers was 2 760 087.

Assuming a visualization with points of diameter 0.085 mm, corresponding to a resolution of 300 DPI, at which individual points are still distinguishable, the figure would require dimensions of approximately 65×65 cm. Such a format makes the visualization difficult to use clearly within a standard A4 layout.

Figures 14, 15 and 16 present plots of the density of prime numbers in angular sectors of width 1° , obtained for increasing ranges of numbers. The circle is divided into 360 equal sectors. The y -axis represents the value of the prime number density, while the x -axis corresponds to sector indices from 1 to 360.

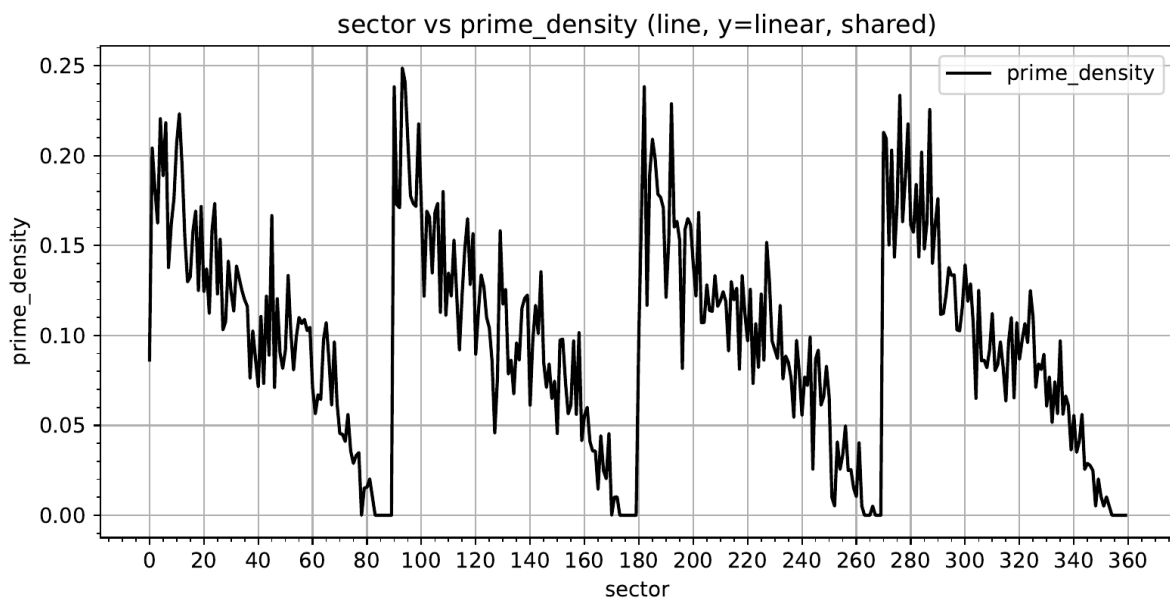


Figure 14: Distribution of the density of prime numbers in angular sectors from 1 to 360 for the following data: number of grid subdivisions $c = 300$; range of numbers 1–70 681; number of cells = 70 681; number of layers = 5 871; number of prime numbers = 7 002. The average density of prime numbers in this range is 9.91% (see Figure 10).

It is observed that as the analyzed numerical range increases, the profile of the density variation becomes progressively smoother. This phenomenon is clearly visible in the RLS visualizations for increasingly large ranges of numbers (see Figures 10, 11, 12).

Based on the observation of the plot, it can be conjectured that as the analyzed numerical range increases, the variations of the density indicator between successive sectors become progressively smoother and tend toward a quasi-linear profile.

The observed behavior of the density thus suggests the possible existence of regularities in the distribution of prime numbers, which allows one to hypothesize that this distribution is not purely random.

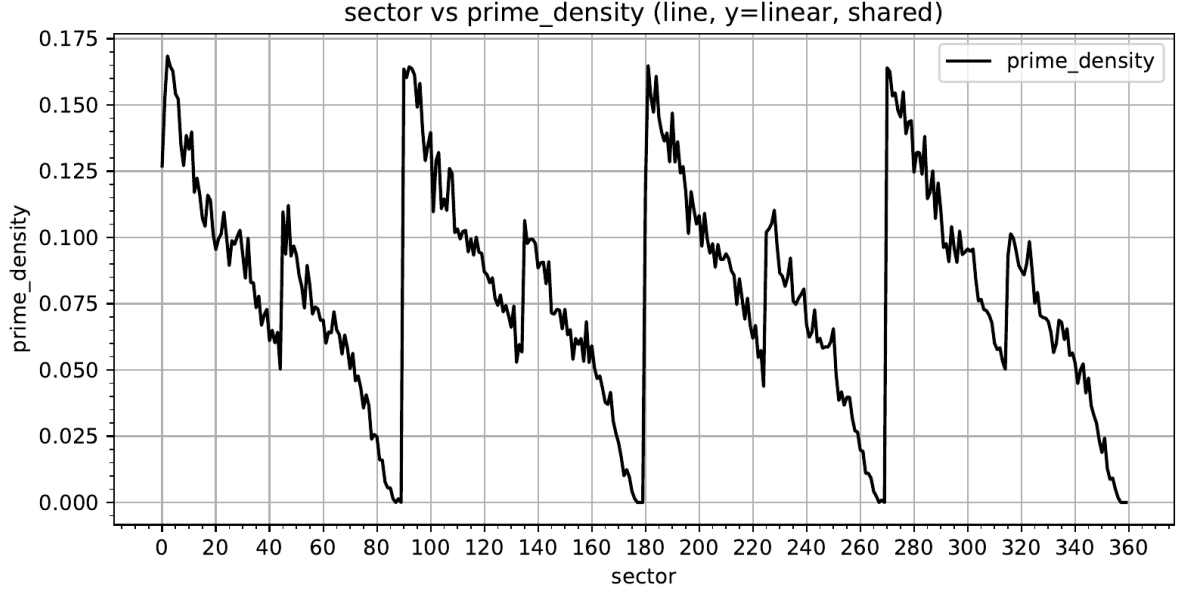


Figure 15: Distribution of the density of prime numbers in angular sectors from 1 to 360 for the following data: number of grid subdivisions $c = 1000$; range of numbers 1–785 349; number of cells = 785 349; number of layers = 57 445; number of prime numbers = 62 869. The average density of prime numbers in this range is 8.01% (see Figure 11).

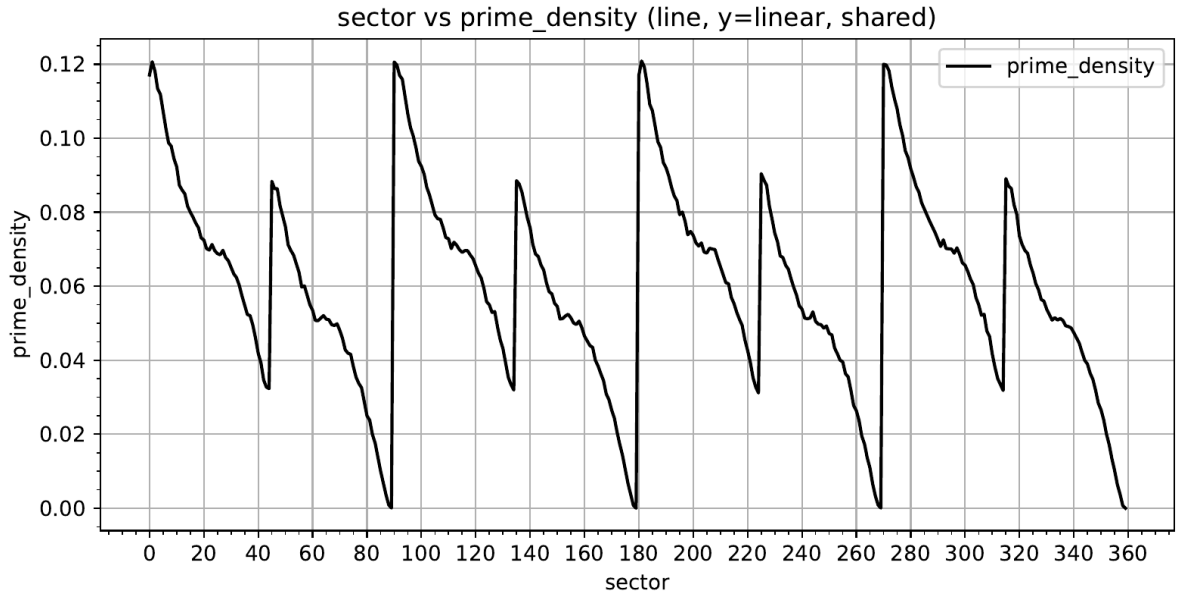


Figure 16: Distribution of the density of prime numbers in angular sectors from 1 to 360 for the following data: number of grid subdivisions $c = 7500$; range of numbers 1–44 178 653; number of cells = 44 178 653; number of layers = 2 760 087; number of prime numbers = 2 671 482. The average density of prime numbers in this range is 6.05%.

4 Analysis of the Layers of the Radial Layer Spiral (RLS)

In the next stage, an analysis of the layers of the Radial Layer Spiral (RLS) and their associated characteristics was carried out. The construction of the layers is illustrated in Figure 17.

4.1 Definition of Layers of the Radial Layer Spiral (RLS)

Consider the equation

$$i^2 + j^2 = m, \quad i, j \in \mathbb{Z}, \quad m \in \mathbb{N}.$$

For a fixed natural number m , there exists a finite number of pairs (i, j) satisfying the above equation. The number of such solutions corresponds to the number of cells in a given layer and is denoted by

$$\text{cells}(m).$$

Each value of m for which there exists at least one solution to the equation $i^2 + j^2 = m$ corresponds to a single layer of the Radial Layer Spiral (RLS). These layers are ordered according to increasing values of m , and the index of a layer is denoted by k (stored in the database as `layer`). Let m_k denote the value of m corresponding to the k -th layer. It should be emphasized that the layer index k is not equal to the value of m , but rather represents its index in the ordered set of such values.

4.2 Layer Radius

The radius of a layer in grid units is defined by

$$r_{\text{grid}} = \sqrt{m}.$$

In the geometric model of the RLS, we consider a circle of radius R and a square grid with side length

$$a = 2R, \quad b = \frac{a}{c},$$

where c is an even number of grid subdivisions. The quantity b denotes the side length of a single grid cell.

The radius of a layer in geometric units is therefore

$$r_p = b\sqrt{m}.$$

The quantity r_p is stored in the database as the parameter `rp` and uniquely determines the position of a given layer with respect to the center of the spiral.

4.3 Cumulative Number of Cells

The cumulative number of cells up to the k -th layer is defined as

$$\text{cells_c}(k) = \sum_{\ell=1}^k \text{cells}(m_{\ell}),$$

where m_{ℓ} denotes the value of m corresponding to the ℓ -th layer.

This quantity is equal to the number of lattice points $(i, j) \in \mathbb{Z}^2$ satisfying the inequality

$$i^2 + j^2 \leq m_k.$$

This means that $\text{cells_c}(k)$ represents the exact discrete number of lattice points contained within a circle of radius r_p .

4.4 Interpretation

In summary:

- the equation $i^2 + j^2 = m$ defines a single RLS layer,
- the number of solutions of this equation is equal to the number of cells $\text{cells}(m)$ in the given layer,
- the radius of the layer is given by $r_p = b\sqrt{m}$,
- the cumulative number of cells $\text{cells_c}(k)$ corresponds to the number of lattice points inside a circle of radius r_p .

In this way, the structure of the Radial Layer Spiral integrates into a coherent whole the discrete geometry of the grid, the arithmetic of sum-of-squares equations, and the continuous geometric description in circular space.

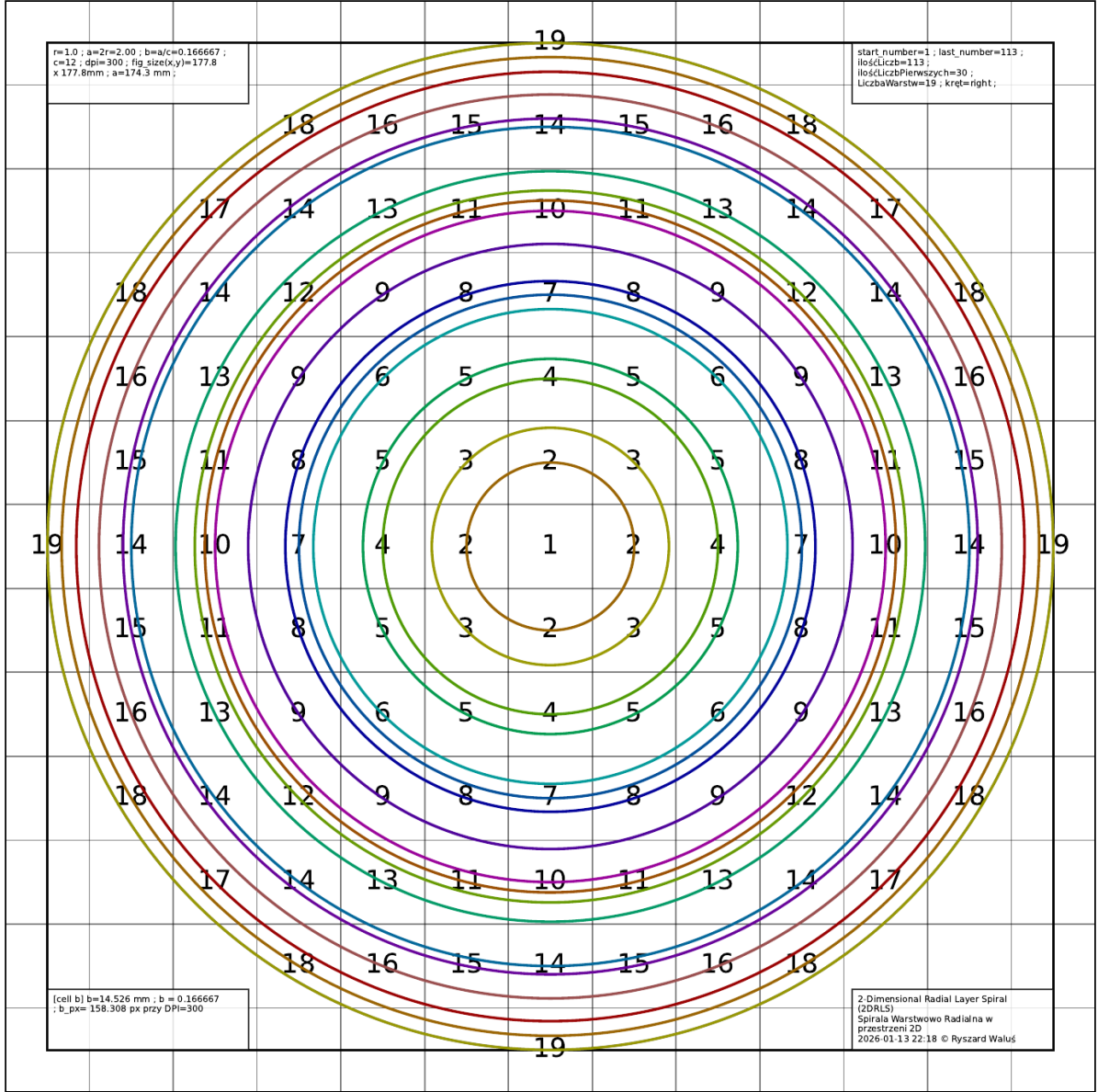


Figure 17: Concentric circles corresponding to successive layers (stored in the database as `layer`) on a square grid. Cells belonging to a given layer are those whose centers lie at the same distance from the origin, so that each circle passes through the centers of the cells forming that layer. Cells are labeled by the index of the layer to which they belong. The distances between successive circles are not constant and result from the discrete nature of the grid, since the admissible radii are of the form $r_p = b\sqrt{i^2 + j^2}$, where $i, j \in \mathbb{Z}$. As a consequence, some radius values occur more densely, while others more sparsely, which leads to irregular spacing between the circles. Figure parameters: number of grid subdivisions $c = 12$; number of points (cell centers) in the circle: 113; number of layers: 19.

layer	m	r_p	r_grid	cells	cells_c
1	0	0.0000	0.0000	1	1
2	1	0.1667	1.0000	4	5
3	2	0.2357	1.4142	4	9
4	4	0.3333	2.0000	4	13
5	5	0.3727	2.2361	8	21
6	8	0.4714	2.8284	4	25
7	9	0.5000	3.0000	4	29
8	10	0.5270	3.1623	8	37
9	13	0.6009	3.6056	8	45
10	16	0.6667	4.0000	4	49
11	17	0.6872	4.1231	8	57
12	18	0.7071	4.2426	4	61
13	20	0.7454	4.4721	8	69
14	25	0.8333	5.0000	12	81
15	26	0.8498	5.0990	8	89
16	29	0.8975	5.3852	8	97
17	32	0.9428	5.6569	4	101
18	34	0.9718	5.8310	8	109
19	36	1.0000	6.0000	4	113

Figure 18: Layers of the RLS (see Figure 17) for $c = 12$ ($b = \frac{1}{6}$). The symbol r_{grid} denotes the layer radius expressed in grid units, with $r_{\text{grid}} = \sqrt{m}$, while $r_p = b r_{\text{grid}}$. The quantity `cells` denotes the number of cells in a given layer, whereas `cells_c` denotes the cumulative number of cells up to that layer.

i	j	m	x	y	r_p	r_grid	n	layer
-5	0	25	-0.8333	0.0000	0.8333	5.0000	70	14
-4	3	25	-0.6667	0.5000	0.8333	5.0000	71	14
-3	4	25	-0.5000	0.6667	0.8333	5.0000	72	14
0	5	25	0.0000	0.8333	0.8333	5.0000	73	14
3	4	25	0.5000	0.6667	0.8333	5.0000	74	14
4	3	25	0.6667	0.5000	0.8333	5.0000	75	14
5	0	25	0.8333	0.0000	0.8333	5.0000	76	14
4	-3	25	0.6667	-0.5000	0.8333	5.0000	77	14
3	-4	25	0.5000	-0.6667	0.8333	5.0000	78	14
0	-5	25	0.0000	-0.8333	0.8333	5.0000	79	14
-3	-4	25	-0.5000	-0.6667	0.8333	5.0000	80	14
-4	-3	25	-0.6667	-0.5000	0.8333	5.0000	81	14

Figure 19: Cells of layer (see Figure 17) `layer` = 14 ($m = 25$, $r_p = \frac{5}{6}$). Layer `layer` = 14 corresponds to the value $m = 25$, which has twelve solutions of the equation $i^2 + j^2 = 25$; these points form a complete, symmetric ring with grid radius $r_{\text{grid}} = 5$ and geometric radius $r_p = 5b$. The table presents the grid coordinates (i, j) and the corresponding geometric coordinates (x, y) . The values x, y are given in geometric units for $R = 1$, with accuracy up to four decimal places. The column `n` contains the consecutive cell indices on the RLS spiral, while `layer` denotes the index of the layer to which a given cell belongs (see Figures 7 and 8).

Figure 20 presents the dependence of the radius r_p on the layer index `layer` for successive layers from 1 to 19 (corresponding to the number of layers, see Figure 17). It is assumed that the radius of the circle bounding the system is $R = 1$. Each layer is assigned a value of the radius r_p , defined as the distance of the center of a cell belonging to a given layer from the center of the circle.

As can be observed in Figure 20, the relation $r_p = f(\text{layer})$ over the small range of layers from 1 to 19 is not linear. The increments of the radius between successive layers are irregular, which is a direct consequence of the discrete nature of the square grid. However, for large ranges of layer indices, this dependence becomes noticeably smoother, as illustrated in Figure 21.

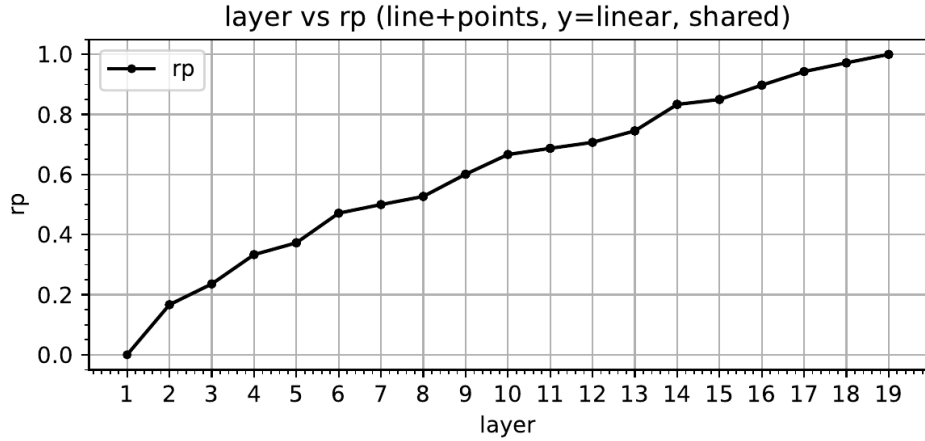


Figure 20: Dependence of the layer radius r_p on the layer index `layer` for a grid with parameter $c = 12$ and the range of layers from 1 to 19, with circle radius $R = 1$.

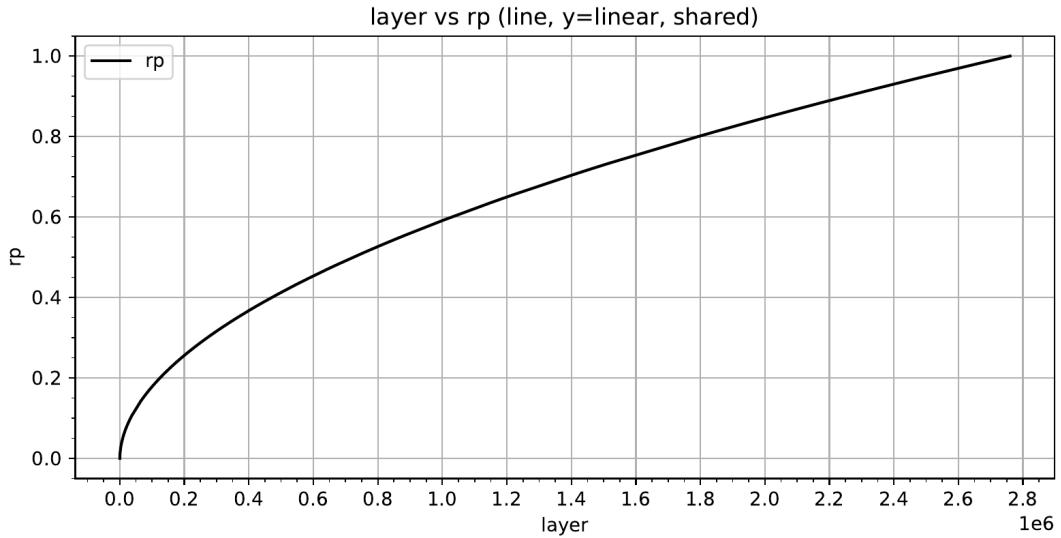


Figure 21: Dependence of the layer radius r_p on the layer index `layer` for a grid with parameter $c = 7500$ and the range of layers from 1 to 2 760 087, with circle radius $R = 1$.

cells	Layers_count	Layer_first	Layer_last
1	1	1	1
4	2474	2	2758661
8	1173564	5	2760086
12	2944	14	2758772
16	1086613	31	2760084
20	199	216	2719001
24	89011	122	2760080
28	37	4170	2706717
32	315733	363	2760079
36	648	1233	2760087
40	3447	2266	2757960
44	1	1940510	1940510
48	45802	1585	2760033
56	174	47135	2754171
60	77	25299	2756775
64	31738	7122	2759970
72	1490	17549	2754167
80	1023	32650	2757013
84	6	549954	2585008
96	4259	37604	2760085
108	15	262065	2396888
112	12	712012	2712664
120	37	379543	2684884
128	655	174927	2759819
144	94	437751	2747256
160	18	821433	2738896
192	15	1198678	2713716

Figure 22: Distribution of the number of cells in the layers of the Radial Layer Spiral (RLS) for a grid with parameter $c = 7500$. The total number of cells is 44 178 653, and the total number of layers is 2 760 087. The column `cells` denotes the number of cells in a layer, while `Layers_count` denotes the number of layers in which a given number of cells occurs. The columns `Layer_first` and `Layer_last` denote, respectively, the first and the last layer in which a given number of cells occurs. The table shows the dominance of layers with 8 and 16 cells, which occur over a wide range of radii, as well as the existence of rare, localized configurations with a larger number of cells, associated with specific arithmetic properties of the equations $i^2 + j^2 = m$.

It is worth noting the row corresponding to the value `cells` equal to 44. In the entire analyzed range, there exists exactly one layer with this number of cells, as indicated by the column `Number_of_layers`. This layer appears for the first (and simultaneously the last) time as layer number 1 940 510, as reflected in the value in the column `Layer_first`.

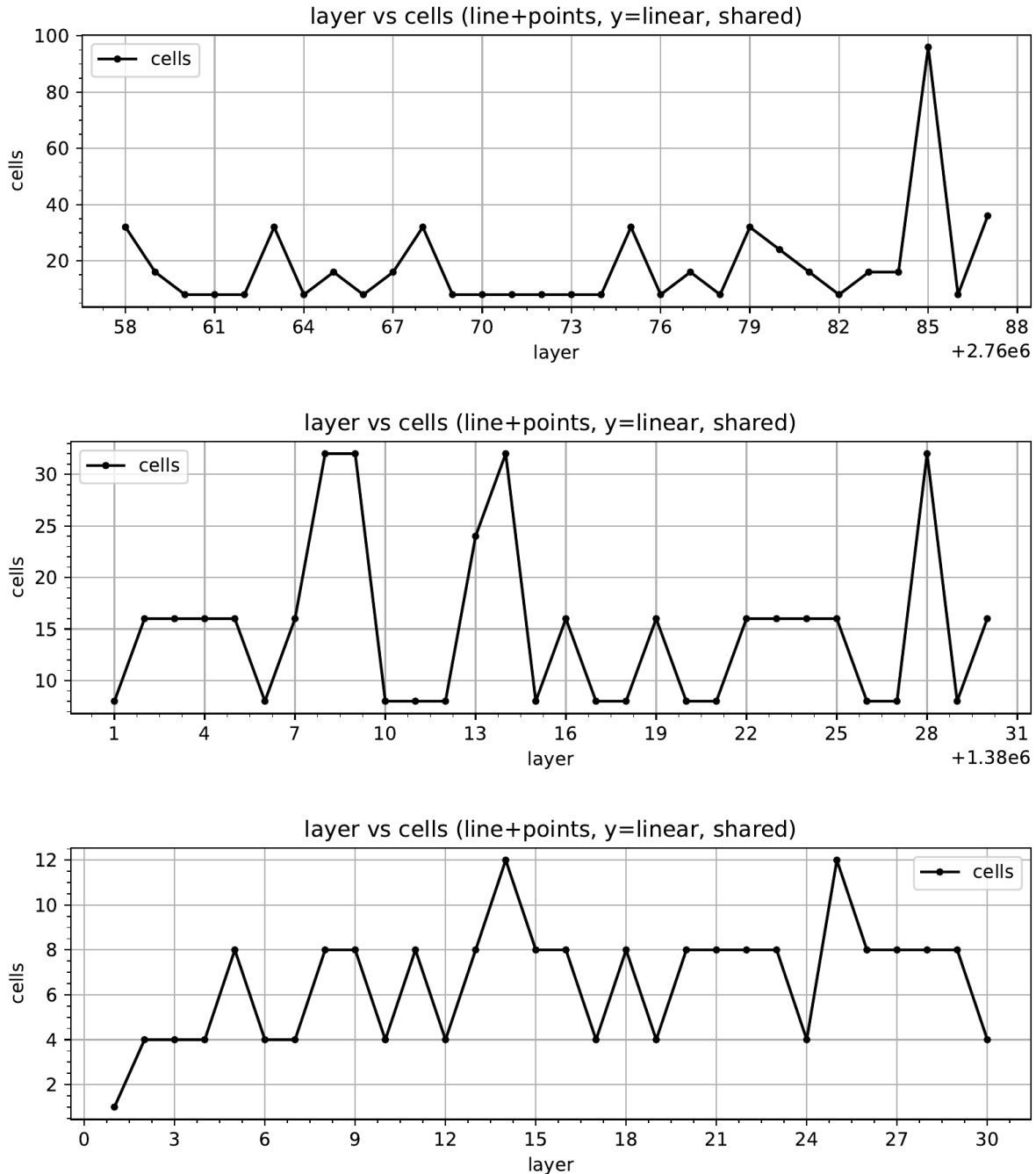


Figure 23: Distribution of the number of cells across individual layers `layer`, illustrated by three segments, each covering 30 consecutive layers. The lower segment represents the range of layers 1–30, the middle segment the range 1 380 001–1 380 030, and the upper segment the range 2 760 058–2 760 087. In the lower range, the minimum number of cells in a layer is 1, and the maximum is 12; in the middle range, these values are 8 and 32, respectively; in the upper range, the minimum number of cells is 8, while the maximum is 96.

The analysis of the table further shows that layers with an increasing number of cells generally appear in increasingly distant regions of the spiral, which is reflected in the growth of the value of `Layer_first` with increasing number of cells. At the same time, the column `Layer_last` indicates that layers with a small number of cells, in particular 4, 8, 12, and 16, occur throughout almost the entire analyzed range of layers.

As each new layer is added, the total number of cells increases by the number of cells belonging to that layer. However, the number of cells in a newly added layer is not arbitrary, but is strictly determined by the number of solutions of the equation

$$i^2 + j^2 = m.$$

It can therefore be concluded that these values exhibit a quantized character, as confirmed by the data listed in the column `cells` in Figure 22. For a grid with parameter $c = 7500$, these values lie in the range from 4 to 192. For larger ranges of the analyzed data, the maximum value of the number of cells in a layer will continue to increase. For example, the value `cells = 160` appears for the first time at `layer = 821 433`, while `cells = 192` occurs starting from `layer = 1 198 678`.

Figure 24 shows the dependence of the total number of cells `cells_c` on the number of layers `layer`. At first glance, this dependence appears approximately linear. However, a more detailed comparison with the linear function

$$f(x) = ax$$

reveals subtle but significant deviations. This comparison is presented in Figure 25, where the dashed line represents the function $f(x) = ax$, which is tangent to the function `cells_c(layer)` at the initial and final points of the analyzed range.

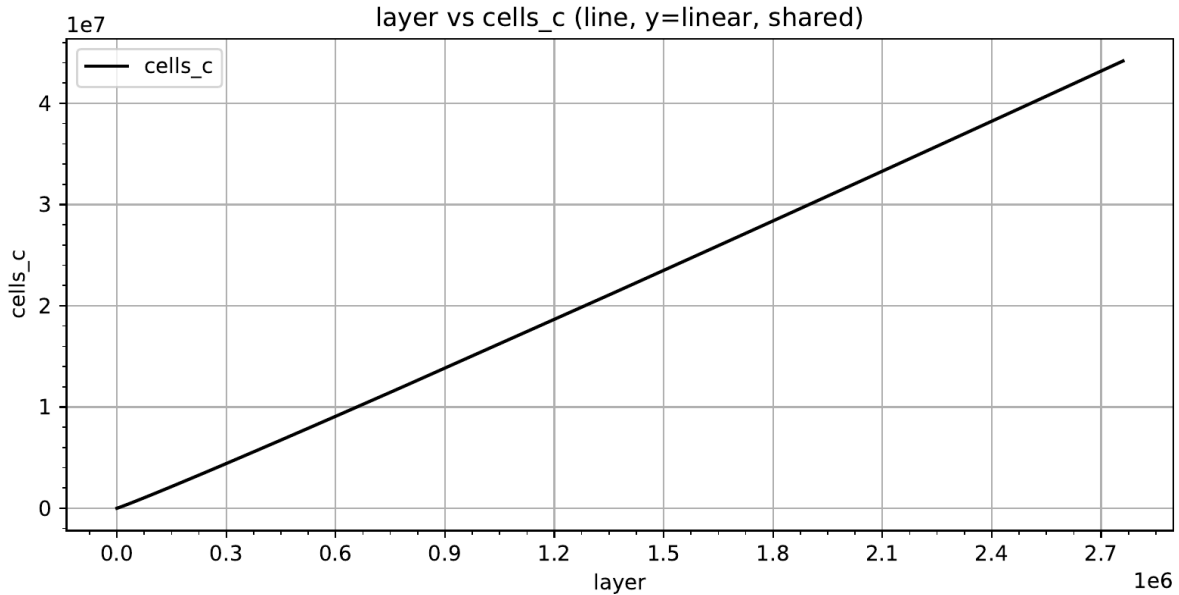
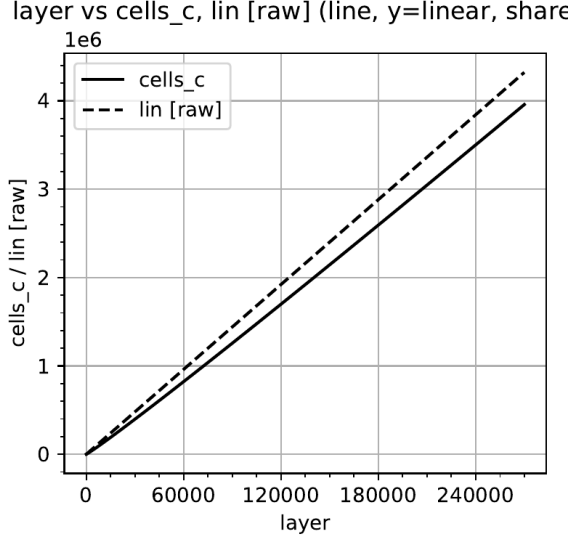
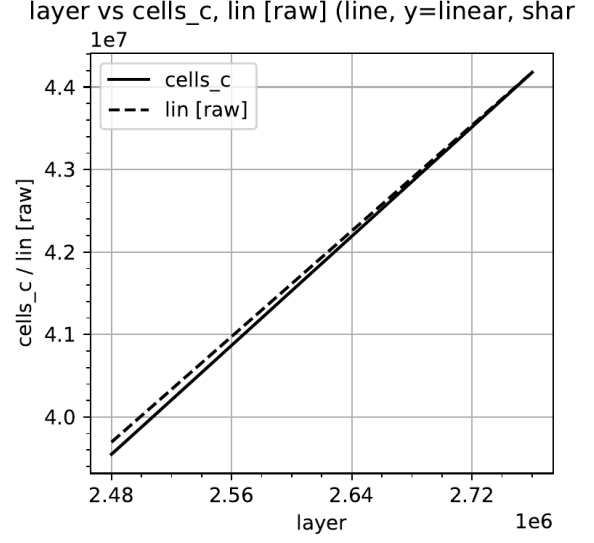


Figure 24: Dependence of the total number of cells (`cells_c`) on the number of layers (`layer`). Parameters of the analyzed grid: $c = 7500$, $R = 1.0$, $b = 0.0002667$, total number of cells = 44 178 653, number of layers = 2 760 087. Y-axis in linear scale.



(a) lower range



(b) upper range

Figure 25: Selected fragments of the plot from Figure 24, covering approximately 10% of the data range from the lower and upper ends, in order to improve the visibility of details. The linear function $f(x) = ax$ is superimposed, where $a = 16.00625$.

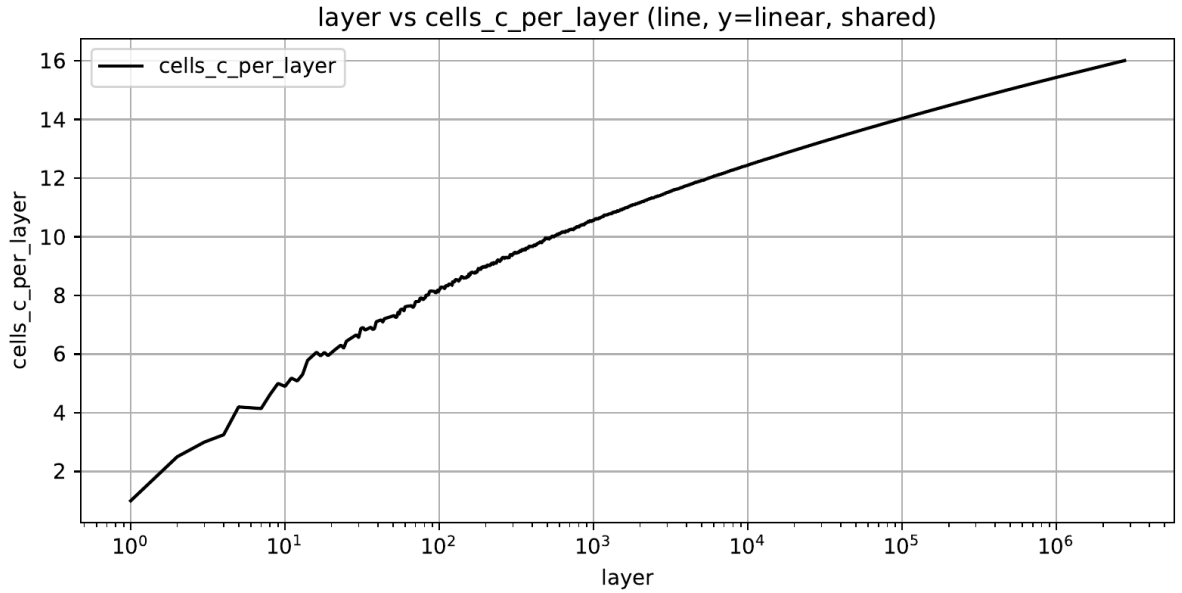


Figure 26: The ratio of the total number of cells to the layer index ($cells_c/layer$) represents the average number of cells per layer. As the number of layers increases, the value of this average also increases. Parameters of the analyzed grid: $c = 7500$, $R = 1.0$, $b = 0.0002667$; total number of cells = 44 178 653, number of layers = 2 760 087. The X-axis is shown on a logarithmic scale.

As can be observed in Figure 26, for very large layer indices, on the order of $layer \approx 2.75 \times 10^6$, the value of the ratio $cells_c/layer$ is approximately 16, which means that in this range the layers contain on average about 16 cells. Similarly, for $layer \approx 10^5$, this average is approximately 14, indicating that in this smaller range the layers contain on average about 14

cells.

From the behavior of the plot, it follows that the ratio $cells_c/layer$ increases with increasing layer index, and therefore the average number of cells per layer exhibits a gradual increase as the number of layers grows. This also implies that successive layers contain, on average, an increasing number of cells.

This phenomenon is confirmed by the data presented in Figure 22, in particular in the column `cells`, where a gradual increase in the number of cells in individual layers can be observed as the layer index increases. The cumulative effect of this process leads to the observed increase in the value of $cells_c/layer$.

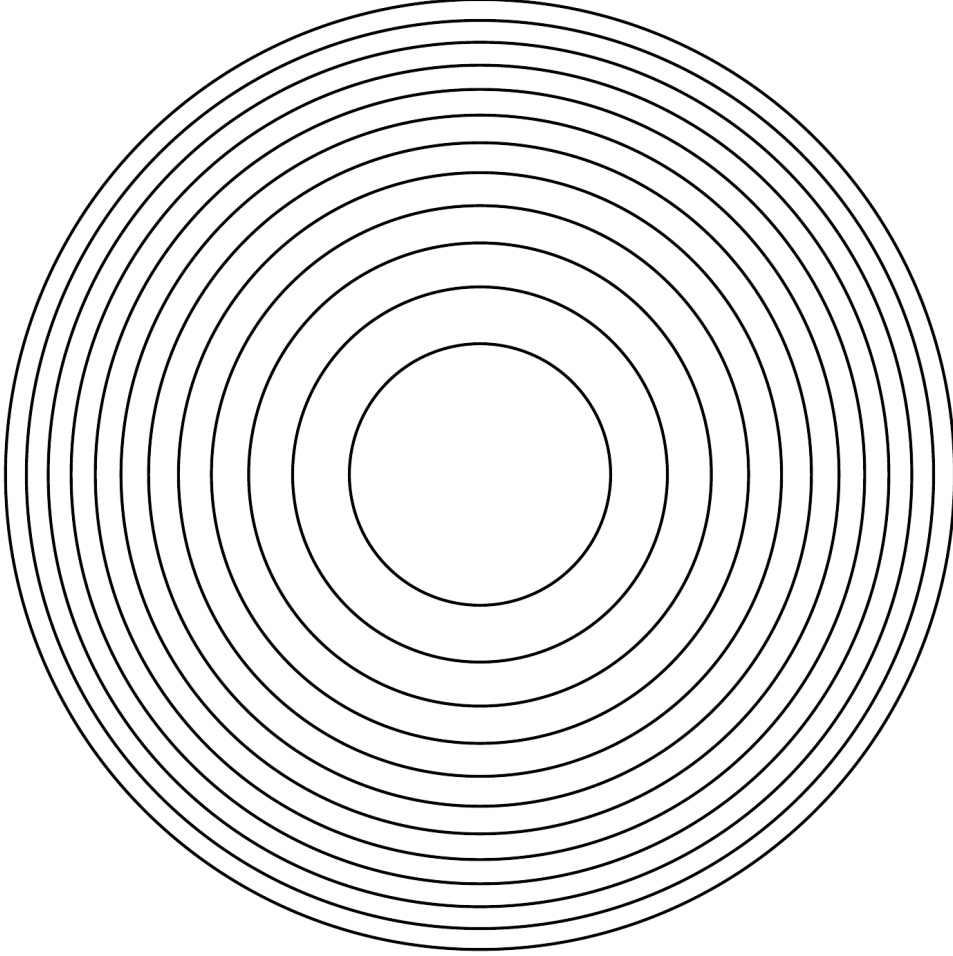


Figure 27: Schematic arrangement of twelve layer sectors. The range of layer indices `layer` is divided into 12 sectors with an equal number of layers. For the data $c = 7500$, total number of cells $N_{cells} = 44\,178\,653$, and total number of layers $N_{layer} = 2\,760\,087$, the width of each sector is $\lfloor N_{layer}/12 \rfloor = 230\,007$ layers.

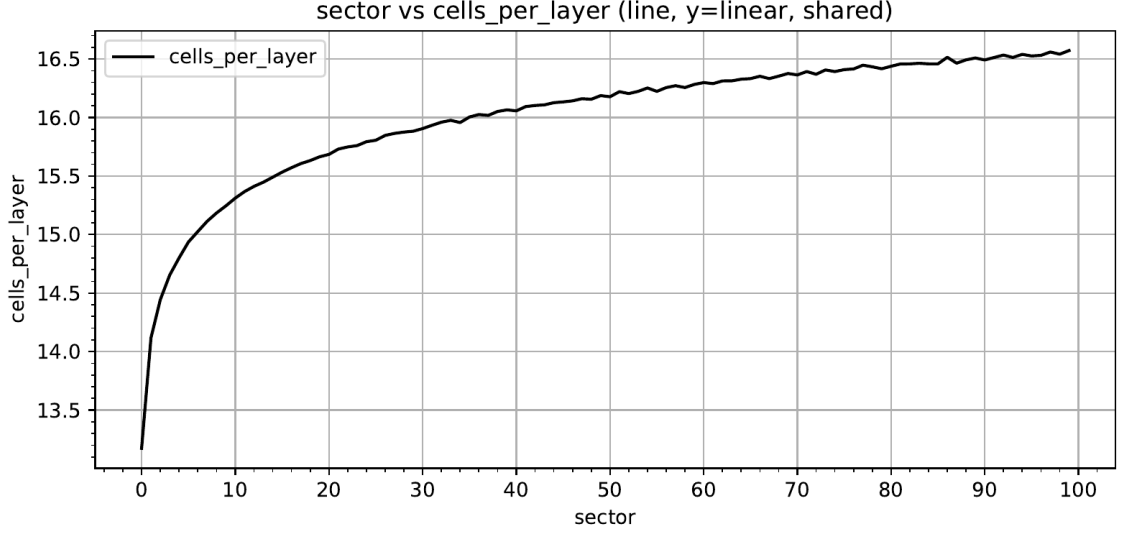


Figure 28: Plot of the average number of cells per layer, computed independently for each layer sector. The circle is divided into 100 layer sectors with an equal number of layers. The computations were performed for the parameters $c = 7500$, total number of cells $N_{\text{cells}} = 44\,178\,653$, and total number of layers $N_{\text{layer}} = 2\,760\,087$, which corresponds to 27 600 layers in each sector. A comparison with Figure 26 shows that the values of the average number of cells determined for the sectors are locally higher than the values computed over the entire range of layers at the corresponding positions on the X-axis, which further highlights the increasing trend of this quantity. It should be noted, however, that in this plot the X-axis has a linear scale ranging from 0 to 100, while covering the same data interval as Figure 26.

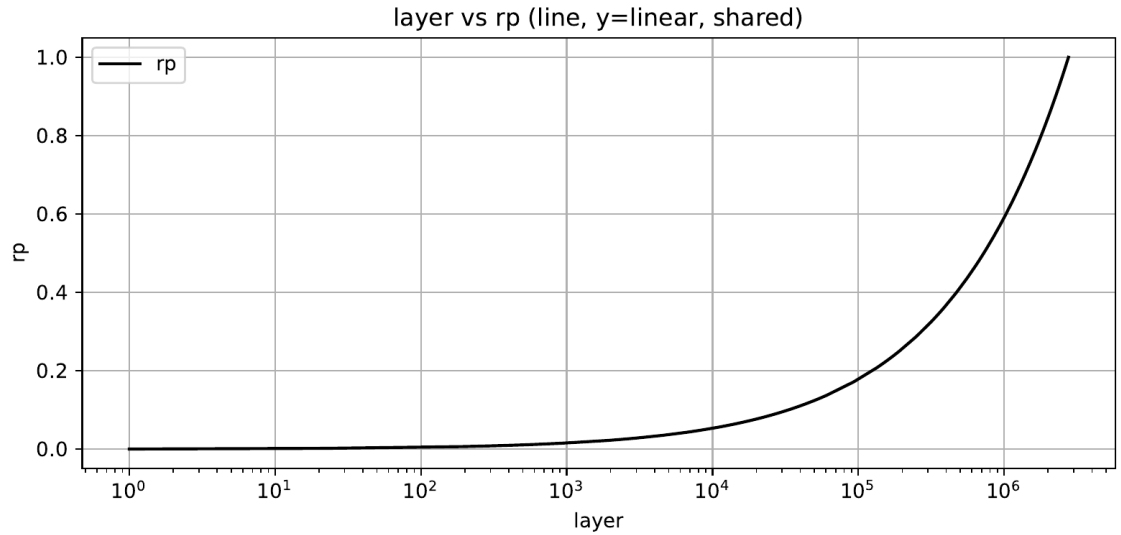


Figure 29: Dependence of the layer radius r_p on the layer index layer . The computations were performed for the parameters $c = 7500$, circle radius $R = 1$, total number of cells $N_{\text{cells}} = 44\,178\,653$, and total number of layers $N_{\text{layer}} = 2\,760\,087$. The plot shows that the density of values of r_p increases with increasing layer index, as illustrated schematically in Figure 27.

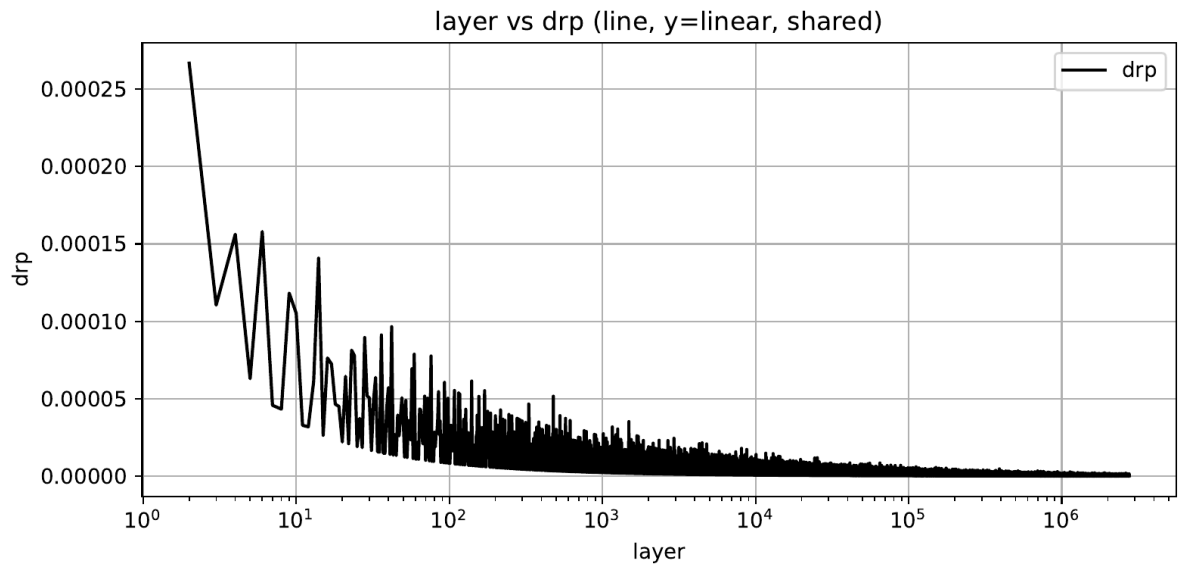


Figure 30: The quantity drp describes the radial distances between successive radii r_p of RLS layers and serves as a measure of the local density of layer circles. Decreasing values of drp indicate an increase in the density of layer circles with radius, while increasing values of drp indicate a decrease in density (see Figures 27 and 29).

5 Axial Cross-Section of the RLS

Consider the axial cross-section of the Radial Layer Spiral (RLS), corresponding to the condition $i = 0$. Then the layer index satisfies the relation

$$m = i^2 + j^2 = j^2,$$

where $i, j \in \mathbb{Z}$ are the grid coordinates of a cell.

The layer radius in grid units is, by definition,

$$r_{\text{grid}} = \sqrt{m} = |j|.$$

Since in the considered range we assume $j \geq 0$, we obtain the simplified form

$$r_{\text{grid}} = j.$$

For successive values of the index j , the grid radius increases linearly, and its increment between adjacent layers is constant and equal to

$$\Delta r_{\text{grid}} = r_{\text{grid}}(j) - r_{\text{grid}}(j - 1) = 1.$$

The normalized radius r_p , which relates the grid radius to the radius of the circle bounding the system, is given by

$$r_p = j \Delta,$$

where

$$\Delta = \frac{2}{c}$$

denotes the normalization step of the radius, and c is the number of grid subdivisions along the diameter of the system.

Hence, the increment of the normalized radius between successive layers for the axial cross-section is

$$\Delta r_p = r_p(j) - r_p(j - 1) = \Delta = \frac{2}{c}.$$

It follows that the difference of radii

$$d_{r_p} = r_p(l) - r_p(l - 1)$$

is constant for the axial cross-section $i = 0$, and the RLS layers are, in this case, equally spaced in the radial direction.

Figure 31 presents values computed for the axial cross-section of the Radial Layer Spiral (RLS), corresponding to the condition $i = 0$. The columns list both geometric quantities and the corresponding parameters of the numbering along the number line.

i Grid coordinate i . In the considered case, it takes the constant value $i = 0$, which means that an axial cross-section of the system is considered (see Figure 32).

j The second grid coordinate $j \in \mathbb{Z}$, ordering successive cells moving axially away from the center of the system; in the analyzed range, $j > 0$ is assumed (see Figure 32).

i	j	n	layer	rp	d_n	d_lay	d_rp
0	1	3	2	0.000267			
0	2	11	4	0.000533	8	2	0.000267
0	3	27	7	0.000800	16	3	0.000267
0	4	47	10	0.001067	20	3	0.000267
0	5	73	14	0.001333	26	4	0.000267
0	6	111	19	0.001600	38	5	0.000267
0	7	147	24	0.001867	36	5	0.000267
0	8	195	30	0.002133	48	6	0.000267
0	9	251	37	0.002400	56	7	0.000267
0	10	309	44	0.002667	58	7	0.000267
0	11	375	52	0.002933	66	8	0.000267
0	12	439	59	0.003200	64	7	0.000267
0	13	521	69	0.003467	82	10	0.000267
0	14	611	78	0.003733	90	9	0.000267
0	15	701	87	0.004000	90	9	0.000267
0	16	795	98	0.004267	94	11	0.000267
0	17	893	109	0.004533	98	11	0.000267
0	18	1007	121	0.004800	114	12	0.000267
0	19	1127	133	0.005067	120	12	0.000267
0	20	1249	146	0.005333	122	13	0.000267
0	21	1371	158	0.005600	122	12	0.000267

Figure 31: Axial cross-section of the RLS for $i = 0$. The table presents the first 21 rows of the dataset, corresponding to computations performed for the parameters: number of grid subdivisions $c = 7500$, circle radius $R = 1$, total number of cells $N_{\text{cells}} = 44\,178\,653$, and total number of layers $N_{\text{layer}} = 2\,760\,087$.

n The index of the successive natural number assigned to a given cell of the RLS spiral. The increase of n is not linear, but follows from the geometry of the spiral as it passes through successive layers (see Figure 33).

layer The index of the RLS layer to which a given cell belongs. Since the layers have a geometric character (layer circles), the increment of the layer index is not constant (see Figure 34).

rp The normalized layer radius, defined by the relation

$$r_p = j \Delta, \quad \Delta = \frac{2}{c}.$$

This quantity describes the radial distance of a cell from the center of the system in units normalized with respect to the circle radius.

d_n The increment of the natural number index between successive values of j , given by

$$\Delta n = n(j) - n(j - 1).$$

This quantity indicates how many consecutive natural numbers occur between adjacent points of the axial cross-section.

d_{lay} The increment of the RLS layer index:

$$\Delta \text{layer} = \text{layer}(j) - \text{layer}(j - 1).$$

It represents the number of layers between successive axial points and increases with distance from the center of the system.

d_{rp} The increment of the normalized radius:

$$\Delta r_p = r_p(j) - r_p(j - 1).$$

For the axial cross-section $i = 0$, this quantity takes a constant value

$$\Delta r_p = \frac{2}{c},$$

which indicates that the layers are uniformly spaced in the radial direction.

The analysis of the table leads to the following conclusions.

In the axial cross-section of the RLS model, the normalized radius r_p increases linearly with the index j , and its increment d_{rp} remains constant.

At the same time, the increment of the layer index d_{lay} and the increment of the number index d_n exhibit nonlinear growth, resulting from the layered geometry of the spiral.

This indicates that in the RLS model the geometric structure of the space has a primary character, while the numbering of natural numbers is secondary and adapts to the imposed layered structure.

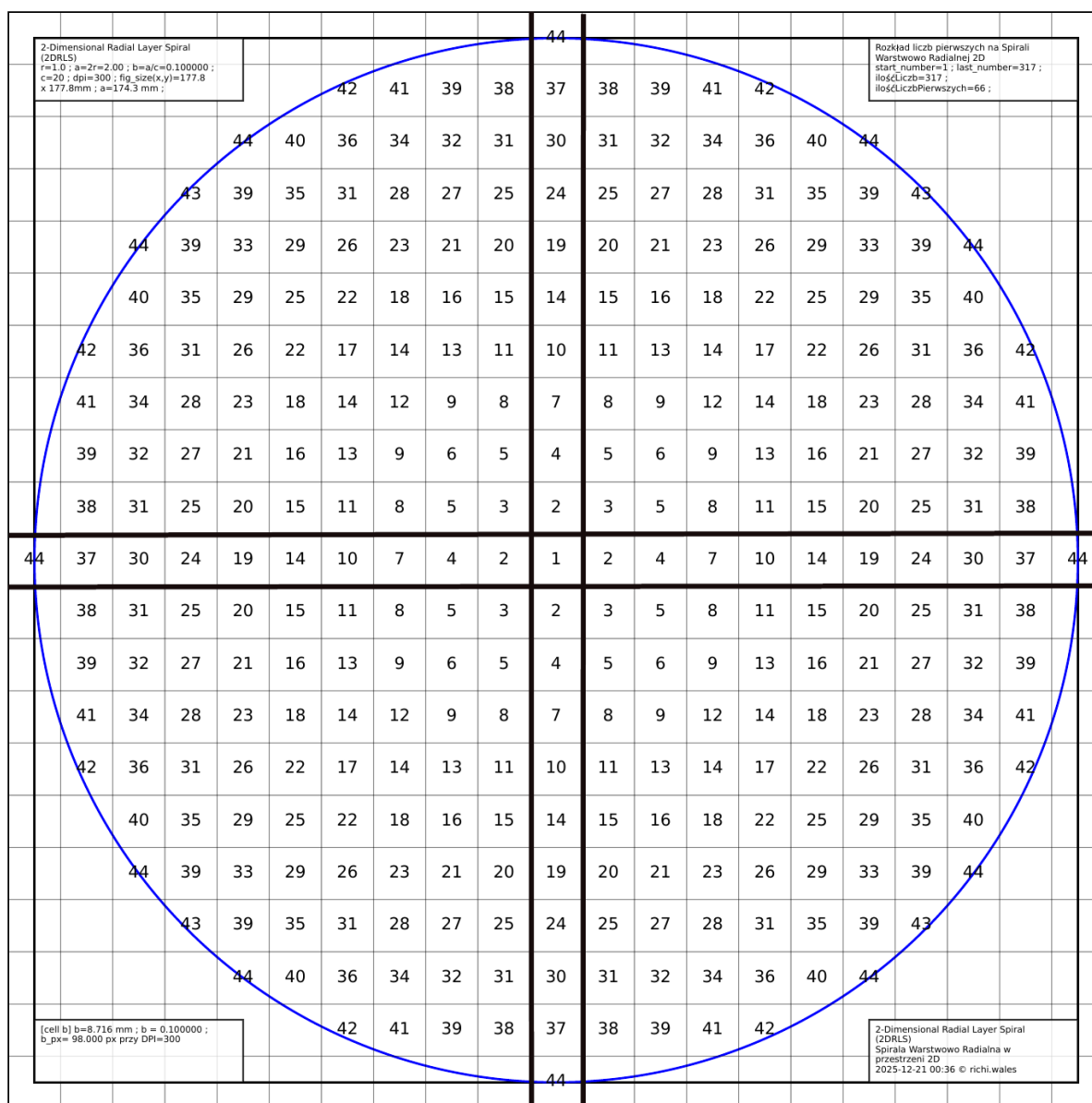


Figure 34: Distribution of layer indices ($layer$) on a square grid. Layers are shown for the range of natural numbers from $n = 1$ to $n = 317$, with grid subdivision parameter $c = 20$. See column $layer$ in Table 31, which lists successive values of layer indices occurring along the Y -axis for the cross-section $i = 0$ and $j > 1$, i.e., for the points (i, j) : 2, 4, 7, 10, 14, 19, 24, 30, 37, 44.

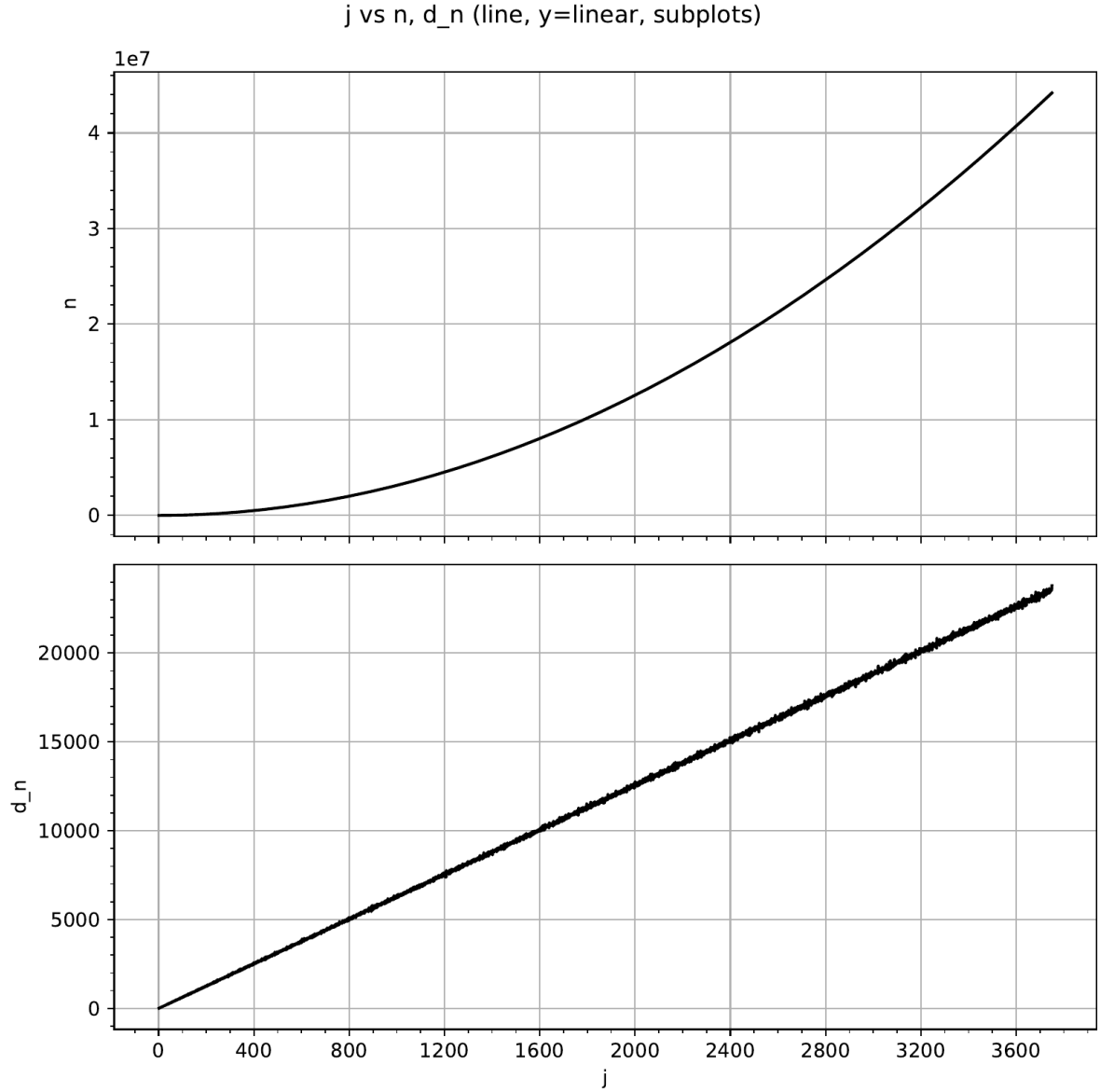


Figure 35: Relations $n = f(j)$ and $d_n = f(j)$ for the axial cross-section of the RLS with $i = 0$ and $j > 0$. The variable n denotes the successive natural number along the RLS axis, while d_n is the increment of the natural number index between successive values of j , defined by $\Delta n = n(j) - n(j - 1)$. RLS data range: $c = 7500$, range of n (1, 44 178 653), range of j (1, 3750). The plot of $d_n = f(j)$ is approximately linear, while exhibiting a clear increase in the amplitude of fluctuations (“noise”), resulting from the increasing differences between successive values of d_n . Details of this effect are shown in Figure 36.

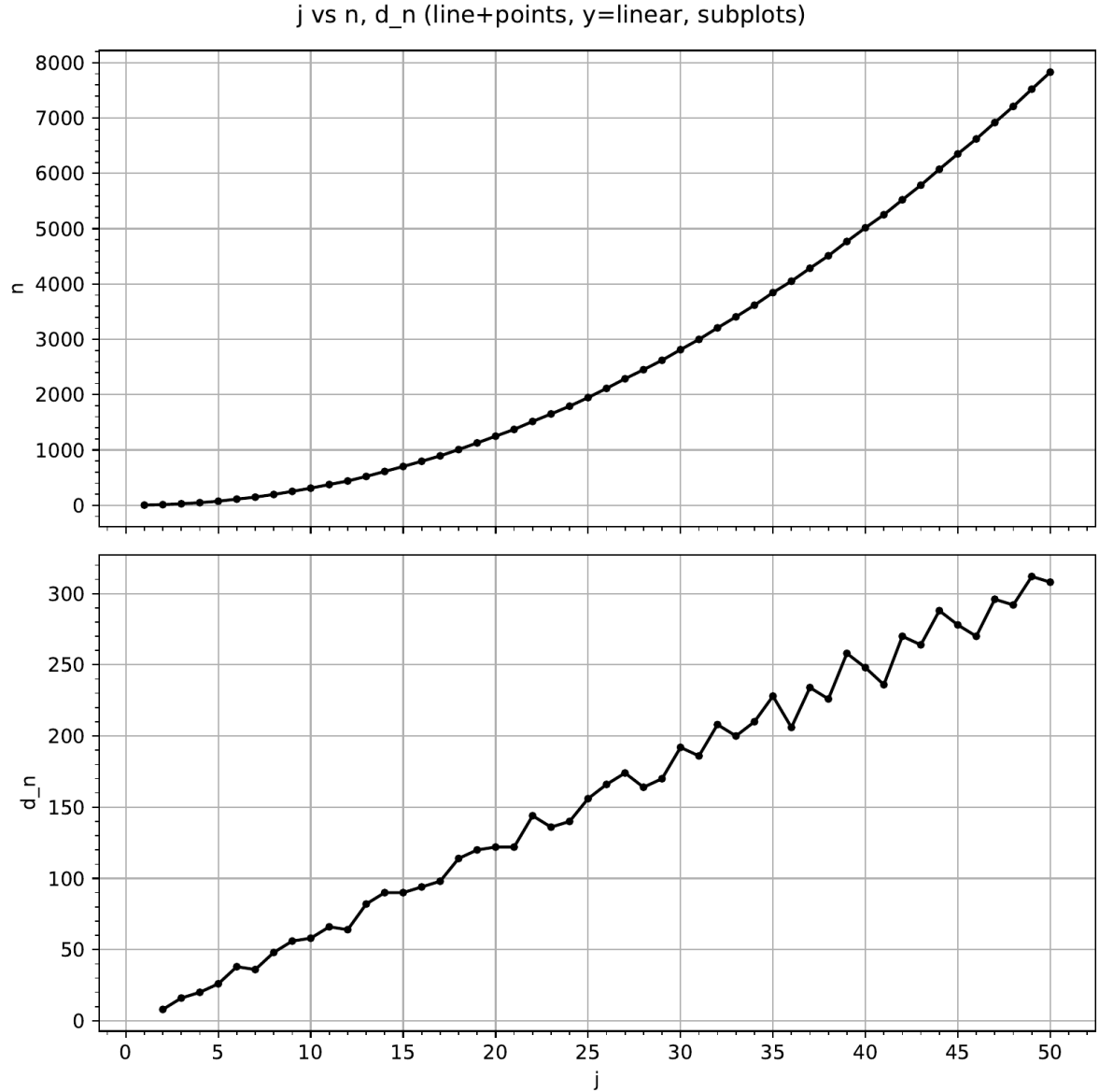


Figure 36: Relations $n = f(j)$ and $d_n = f(j)$ for the axial cross-section of the RLS with $i = 0$ and $j > 0$, shown for the range $j \in (1, 50)$. The function $d_n = f(j)$ forms a sequence of discrete values that approximately fluctuate around a trend curve with a near-linear character. For the narrow range of j presented in this figure, these fluctuations are clearly visible. For a wider range of j (see Figure 35), their relative amplitude is smaller; however, as j increases, the absolute amplitude of the fluctuations grows, as illustrated in Figures 37 and 38.

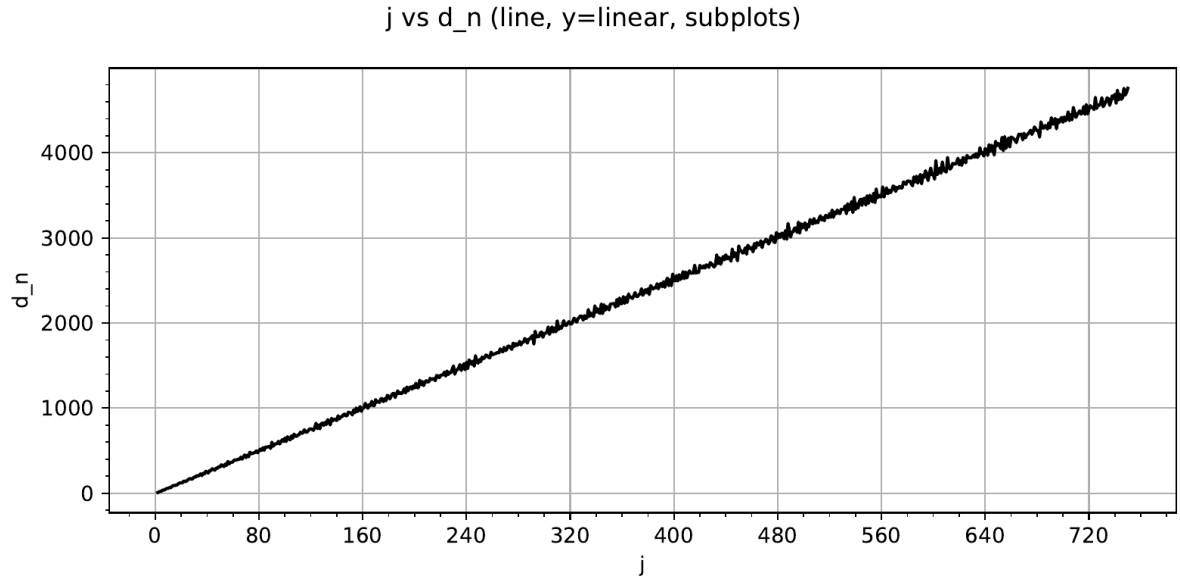


Figure 37: Relation $d_n = f(j)$ for the axial cross-section of the RLS with $i = 0$ and $j > 0$, for the range $j \in (1, 750)$

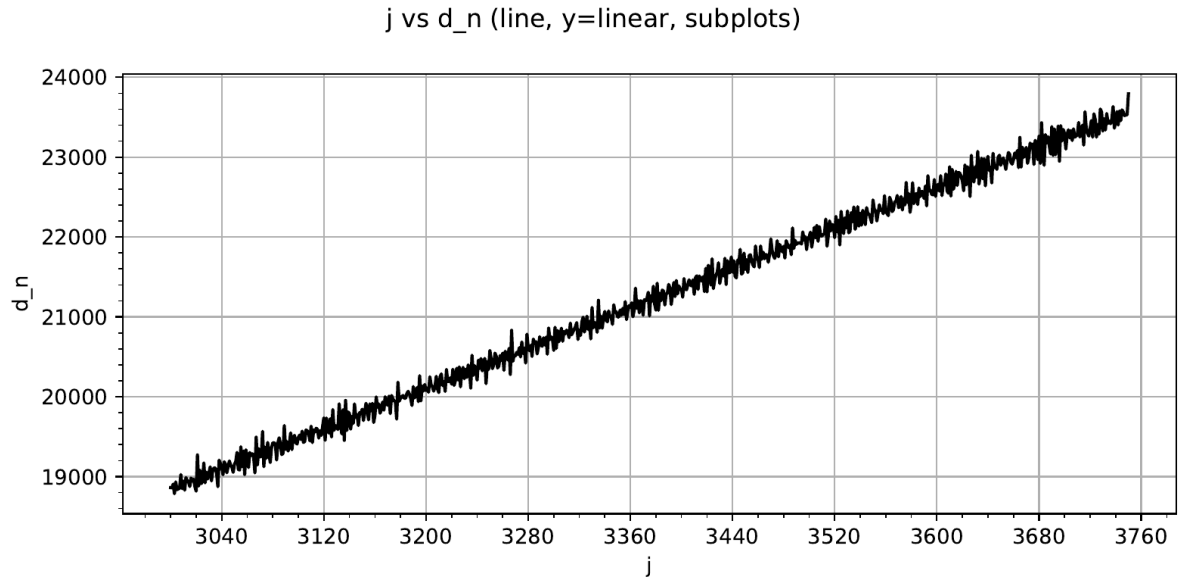


Figure 38: Relation $d_n = f(j)$ for the axial cross-section of the RLS with $i = 0$ and $j > 0$, for the range $j \in (3000, 3750)$

6 Mapping of Natural Numbers in the RLS Structure

6.1 Nature of the Mapping

The mapping characteristic of the RLS,

$$(i, j) \mapsto n,$$

is constructive in nature and does not occur in the form of an explicit function. Consequently, determining the value of n for given coordinates (i, j) requires either executing the algorithm generating the spiral or reading the value from a previously generated data table.

Analogously, the inverse mapping

$$n \mapsto (i, j),$$

also has a constructive character and cannot be reduced to the application of an explicit elementary formula. This means that the RLS structure is not described by a simple analytical relation, but rather results from a process of iterative geometric construction.

6.2 Algorithmic Interpretation

The structure of the RLS spiral may be interpreted as a record of the process of its own spatial construction. If the number n is assigned the role of discrete time, then each successive step of the algorithm corresponds to a uniquely determined state of the system.

Within this interpretation, the RLS enables the reconstruction of the entire previous evolution of the structure on the basis of its current state. Future states of the spiral, however, are not available in explicit form and can only be determined through the execution of subsequent algorithmic steps. In this sense, the future of the RLS is potentially contained within its code, yet remains unresolved until further iterations are carried out. This state may be interpreted metaphorically as “blurred”, which finds visual reflection in Figures 36, 37, and 38.

The structure therefore has a deterministic character; however, successive states can only be determined iteratively.

6.3 Geometric Properties and the Distribution of Prime Numbers

The RLS spiral guides the path of the number axis in a locally unpredictable manner, resulting from the absence of an explicit formula determining the position of the number n on the coordinate grid (i, j) .

At the same time, the manner in which this path is generated reveals an ordered radial distribution of prime number density. The observed effect indicates that the global properties of the distribution of prime numbers, known from asymptotic analysis, undergo a geometric reorganization when the linear number axis is transformed into a radial structure, becoming visible in the form of spatial regularities (cf. Figures 12 and 16).

In this sense, one may speak of a relationship between the distribution of prime numbers and the geometric structure of the circle. This relation manifests itself through the dependence between radial distance and the organization of points on layered circles, leading to a natural connection with the geometry described by the constant π .

The obtained picture indicates that the RLS representation reveals a structure in which the global properties of the distribution of prime numbers enter into a relationship with the geometry of space, revealing regularities invisible in the classical linear representation.

Classical properties of the distribution of prime numbers are described by the relations (cf. [3]; see also [4]):

$$\frac{\pi(n)}{n} \sim \frac{1}{\log n}, \quad p_{k+1} - p_k \sim \log n.$$

These relations describe average properties of the distribution of prime numbers; however, they do not take into account their spatial organization.

In contrast to the asymptotic approach, the RLS representation exposes an ordered distribution of prime number density throughout the entire scale range, organizing fluctuations in a geometric manner. The observed structures preserve their form as the range of n increases, which indicates the existence of a stable spatial organization.

The obtained results suggest that the RLS representation reveals a structure in which the distribution of prime numbers naturally enters into a relationship with the geometry of the circle. The character of this relationship and its potential significance require further investigation.

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